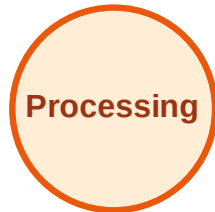


BFS Traversal

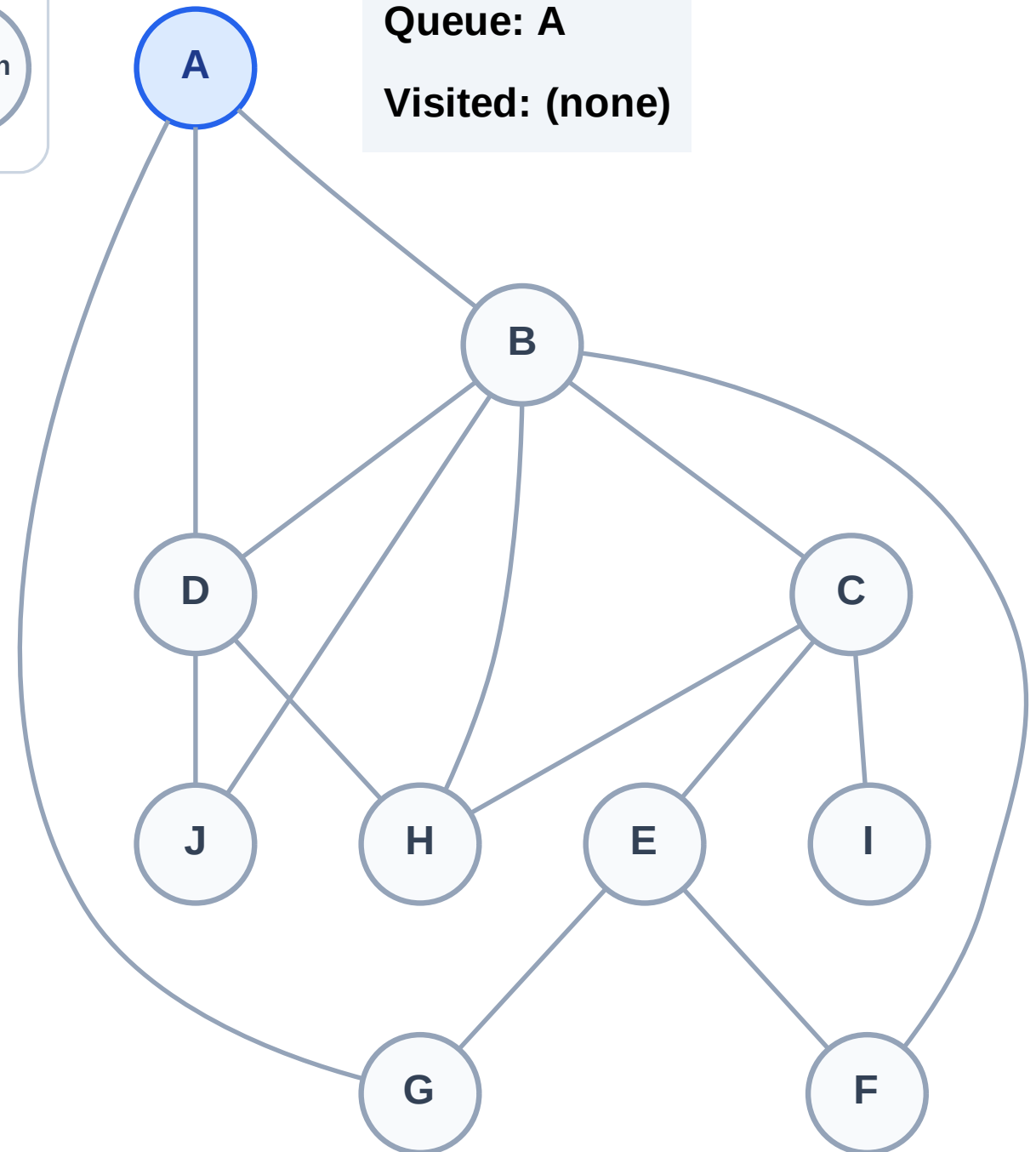
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

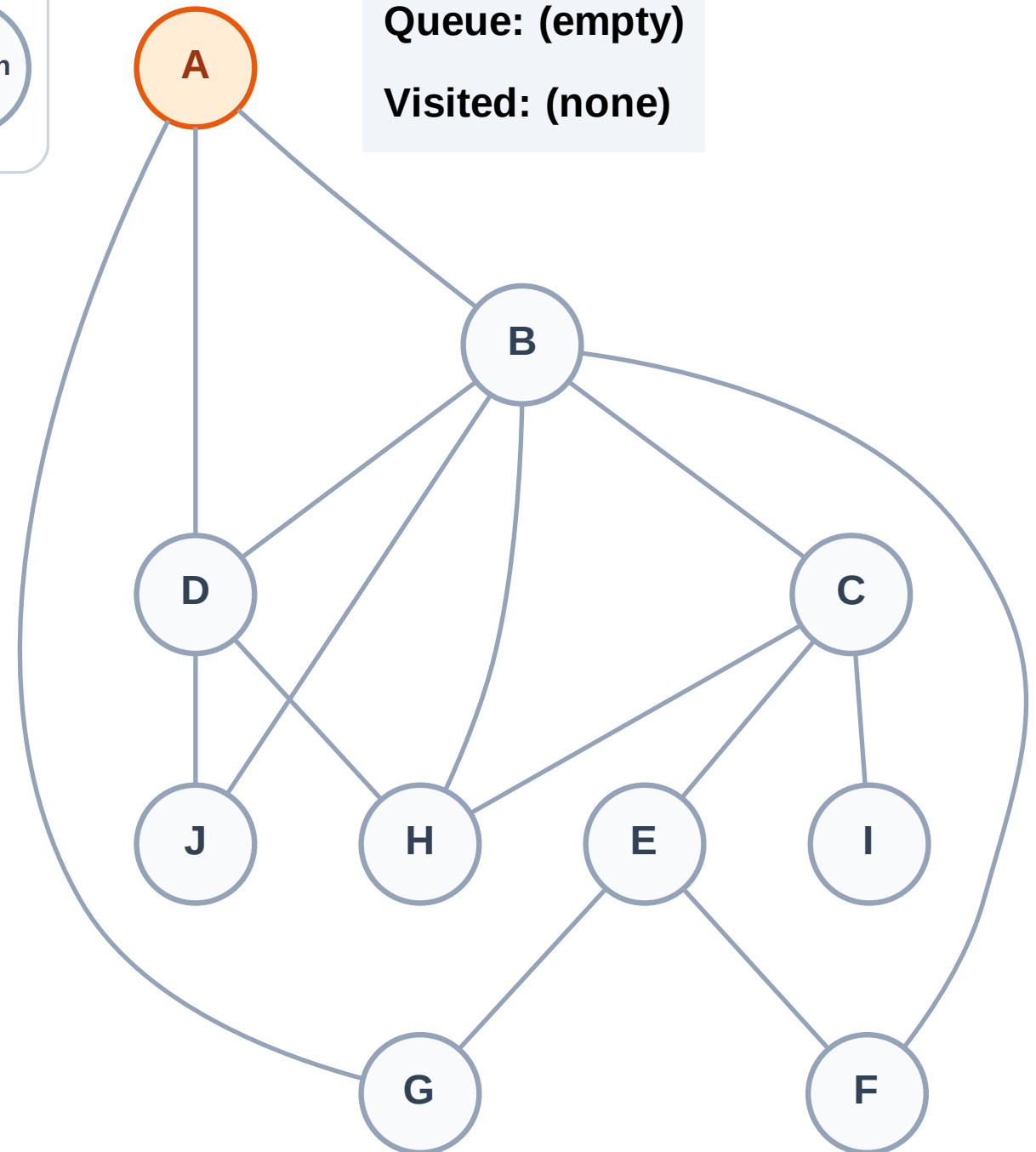
Processing

Discovered

Unseen

Queue: (empty)

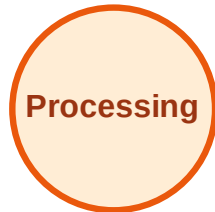
Visited: (none)



BFS Traversal

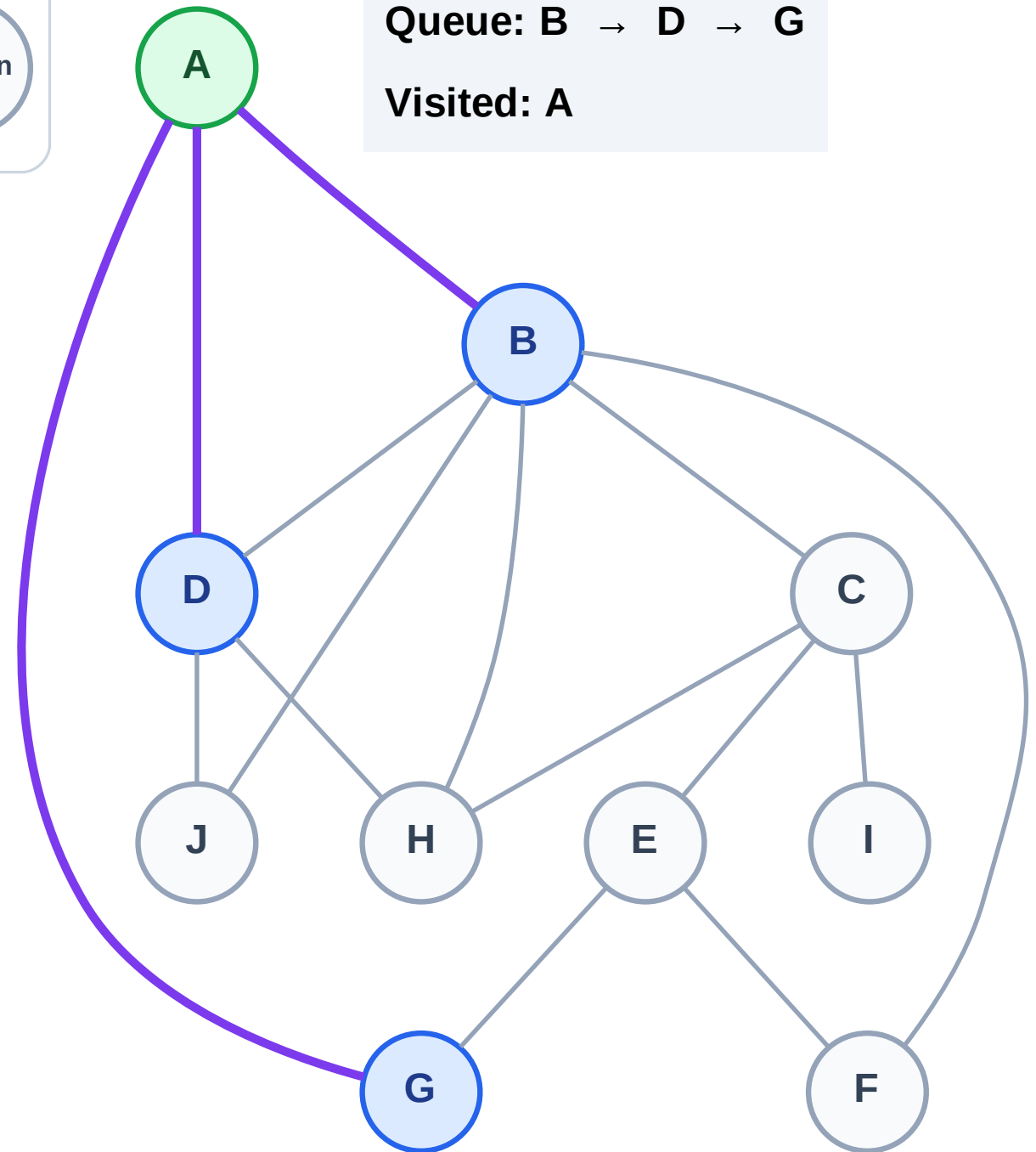
Step 3: Discover B, D, G from A

Legend



Queue: B → D → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

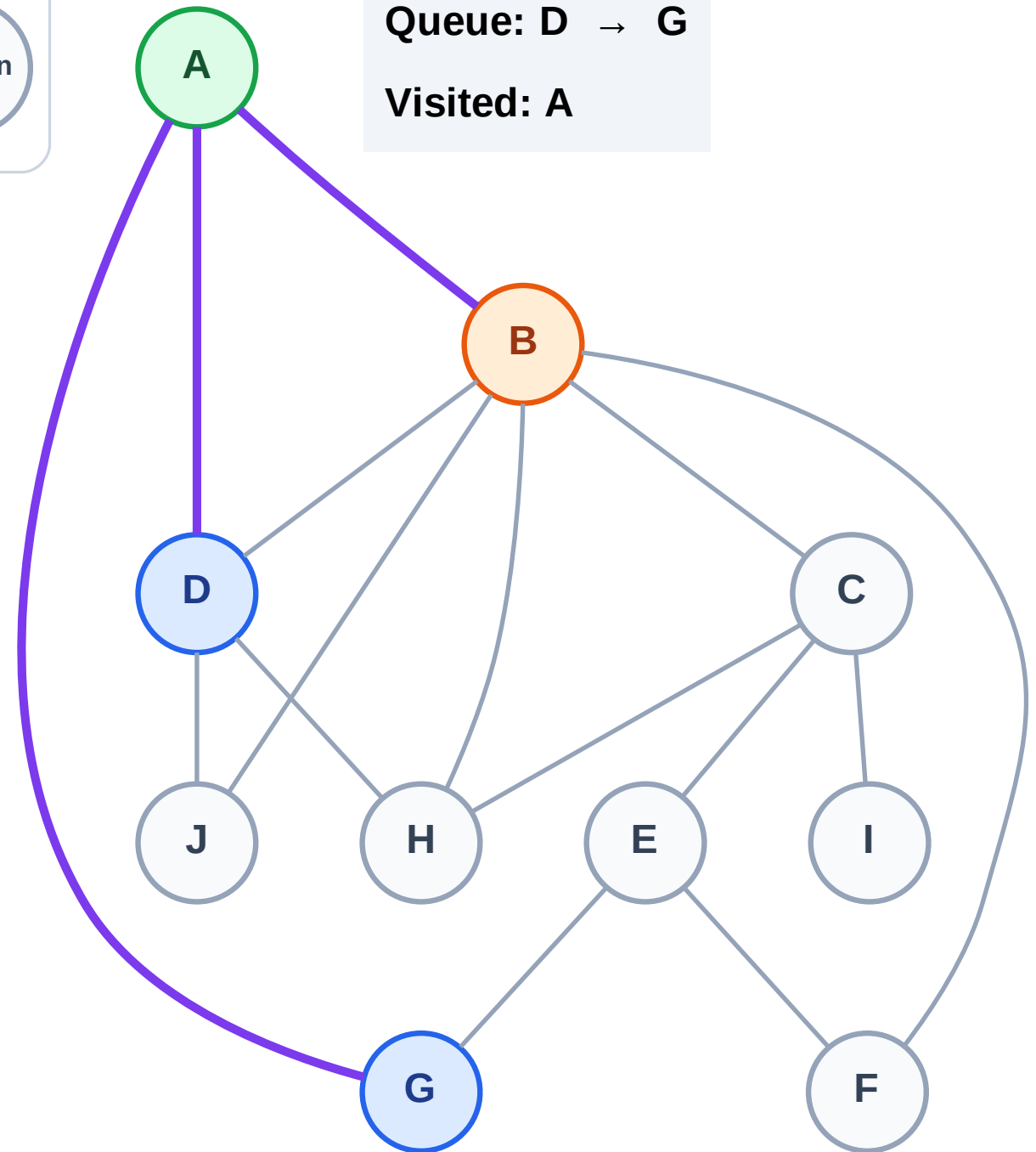
Processing

Discovered

Unseen

Queue: D → G

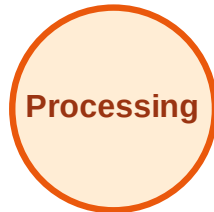
Visited: A



BFS Traversal

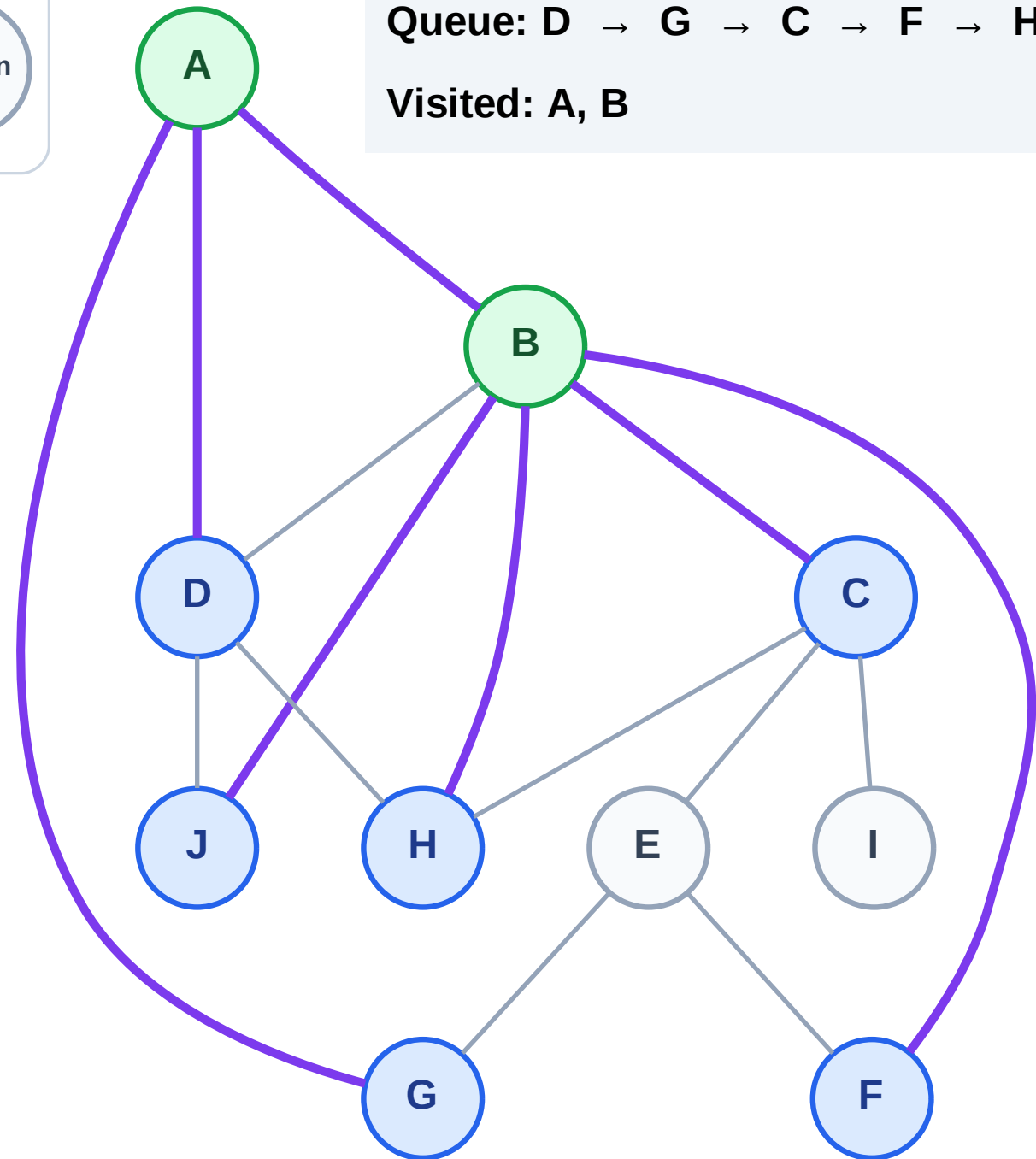
Step 5: Discover C, F, H, J from B

Legend



Queue: D → G → C → F → H → J

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

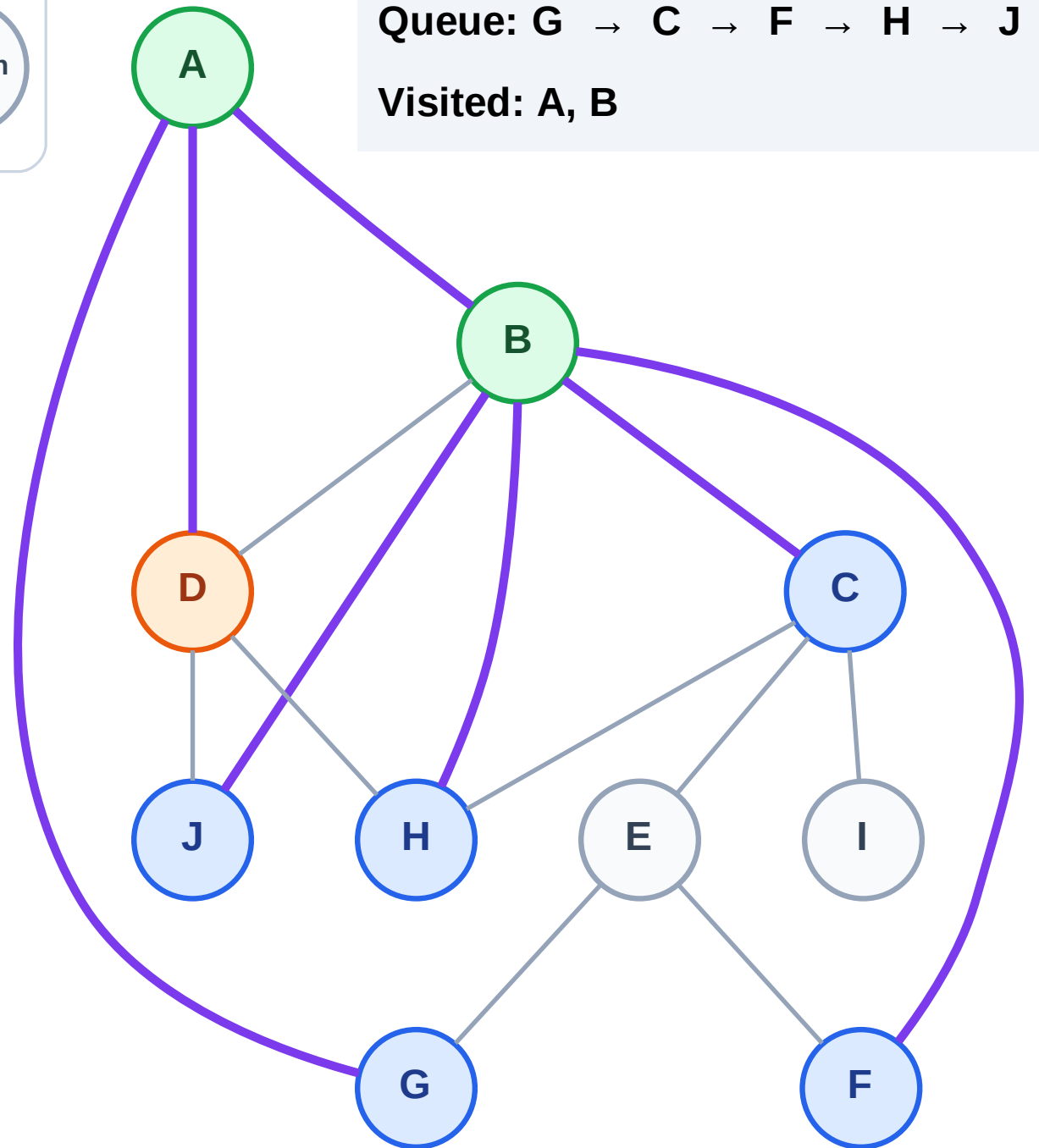
Processing

Discovered

Unseen

Queue: G → C → F → H → J

Visited: A, B



BFS Traversal

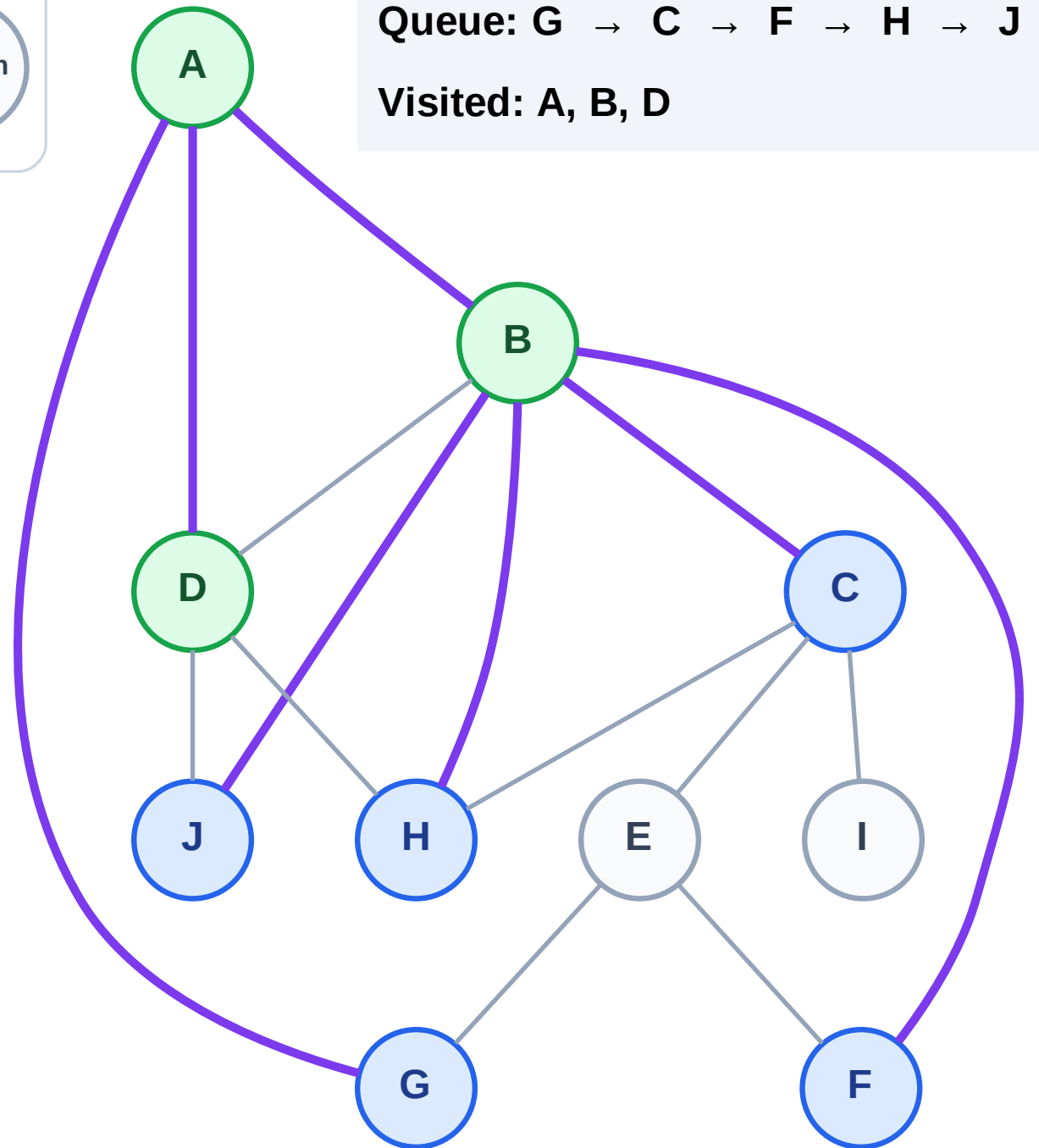
Step 7: No new neighbors from D

Legend



Queue: G → C → F → H → J

Visited: A, B, D



BFS Traversal

Step 8: Process node G

Legend

Complete

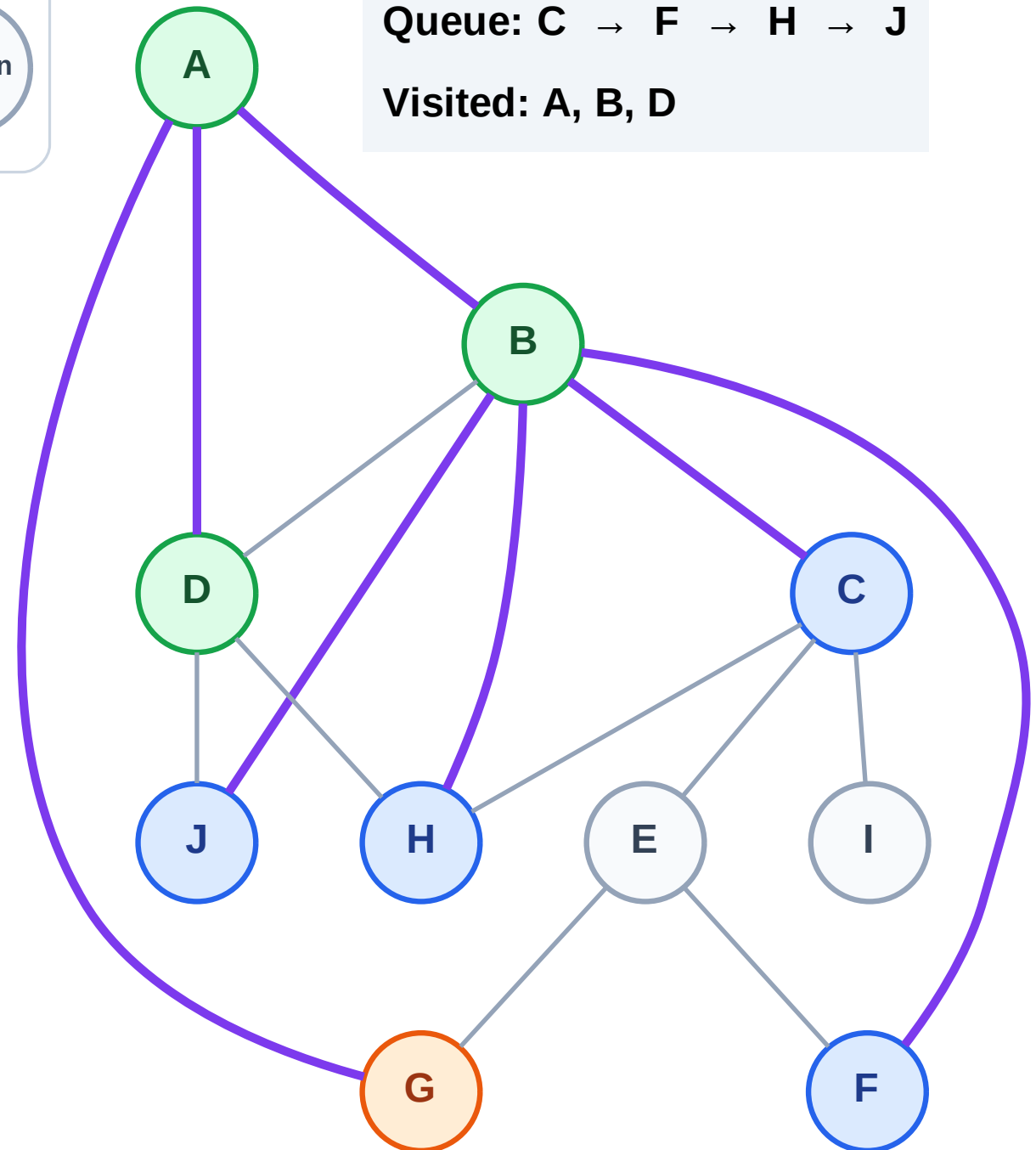
Processing

Discovered

Unseen

Queue: C → F → H → J

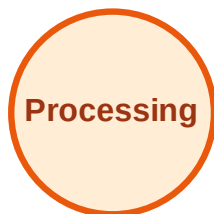
Visited: A, B, D



BFS Traversal

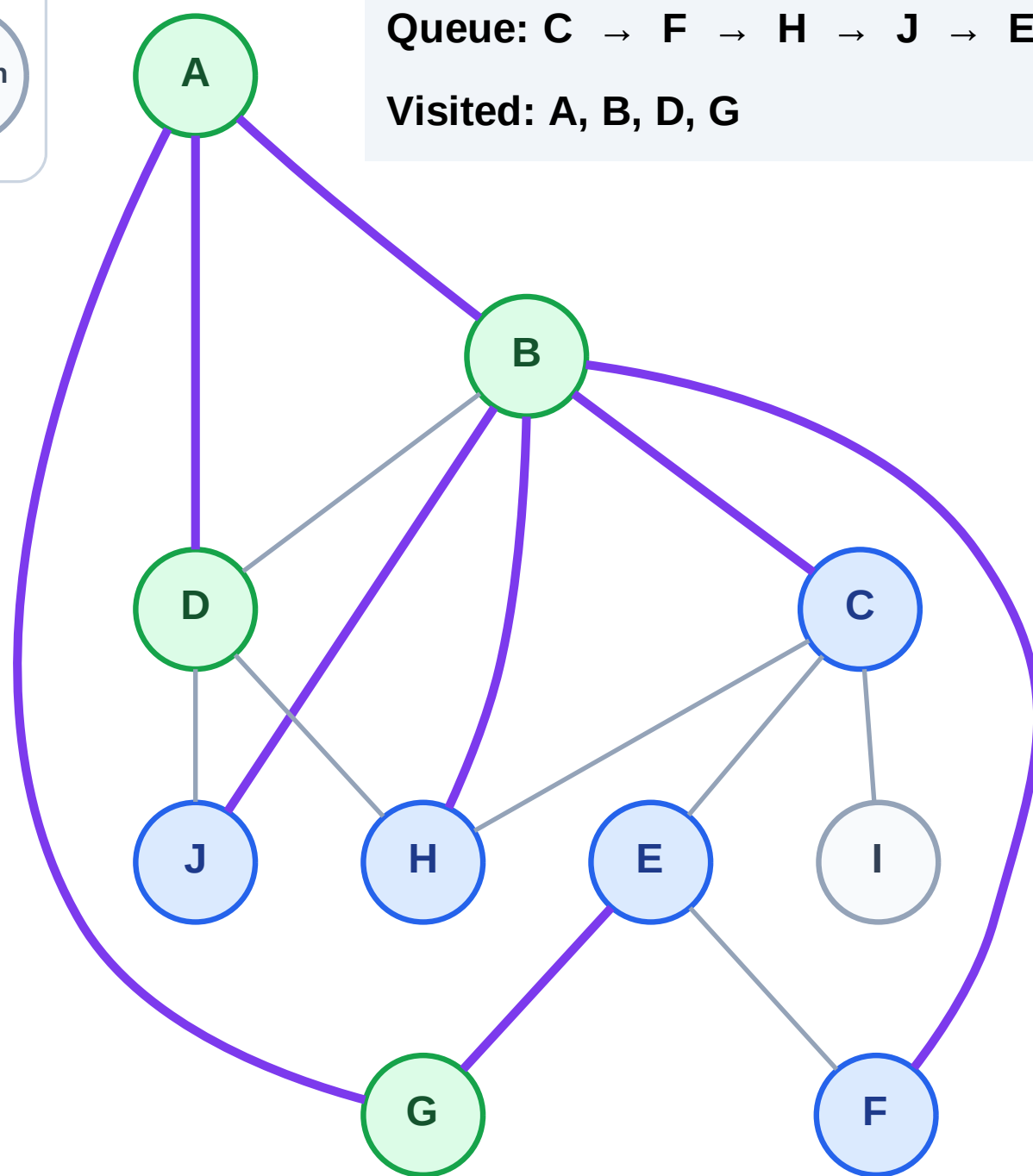
Step 9: Discover E from G

Legend



Queue: C → F → H → J → E

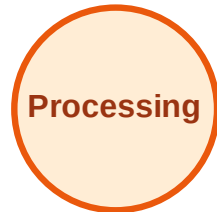
Visited: A, B, D, G



BFS Traversal

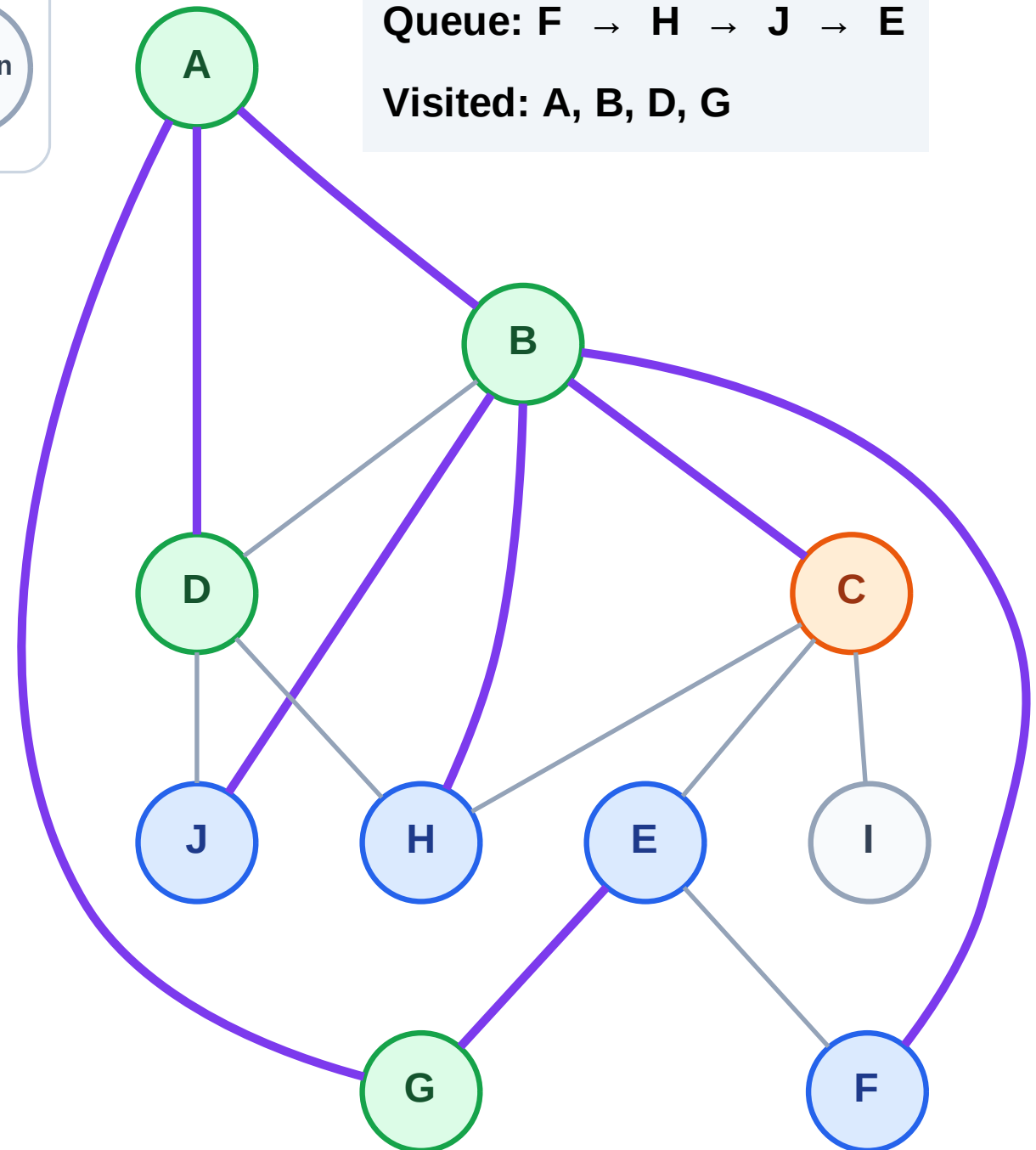
Step 10: Process node C

Legend



Queue: F → H → J → E

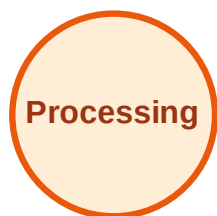
Visited: A, B, D, G



BFS Traversal

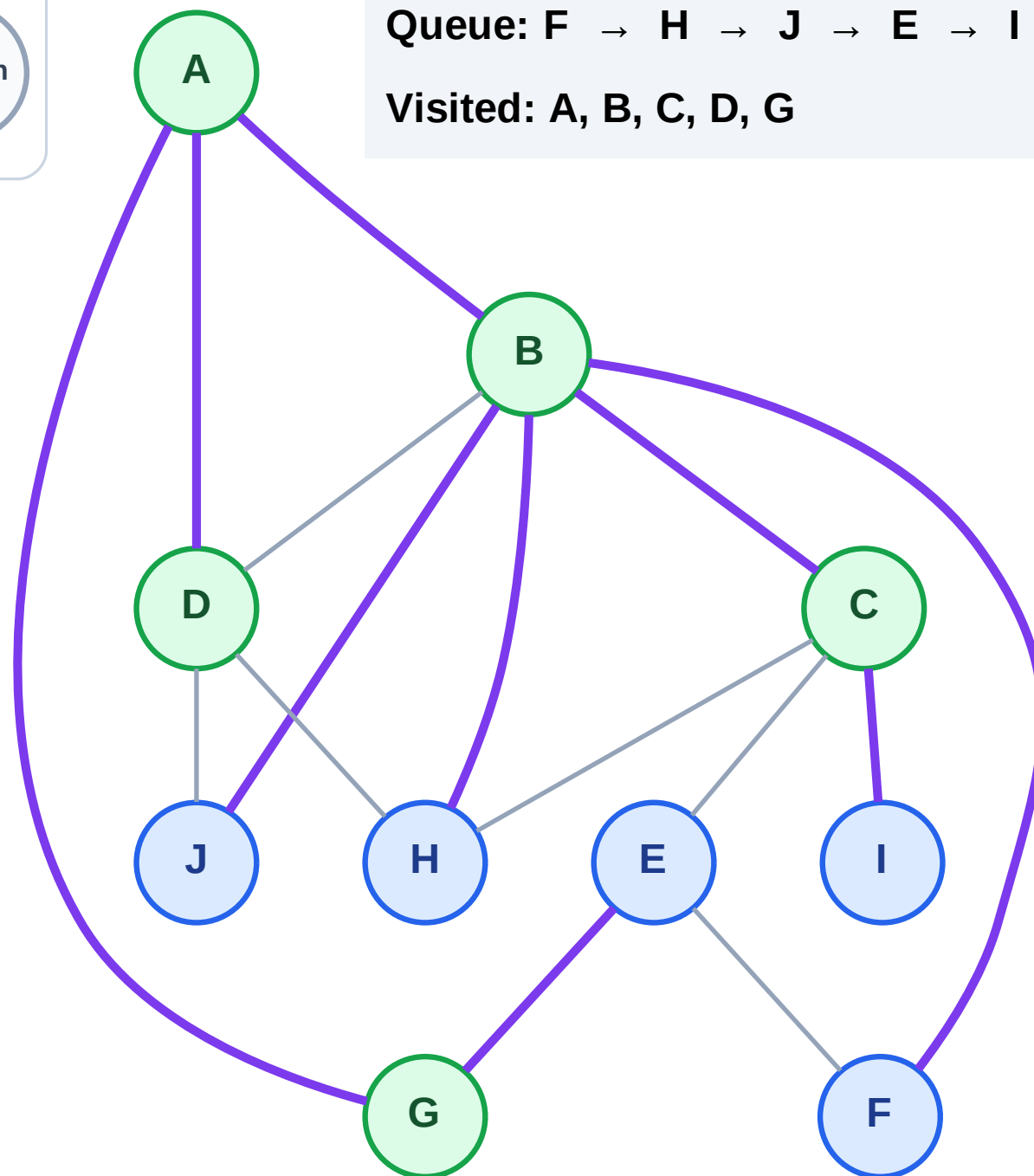
Step 11: Discover I from C

Legend



Queue: F → H → J → E → I

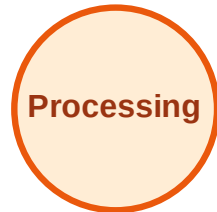
Visited: A, B, C, D, G



BFS Traversal

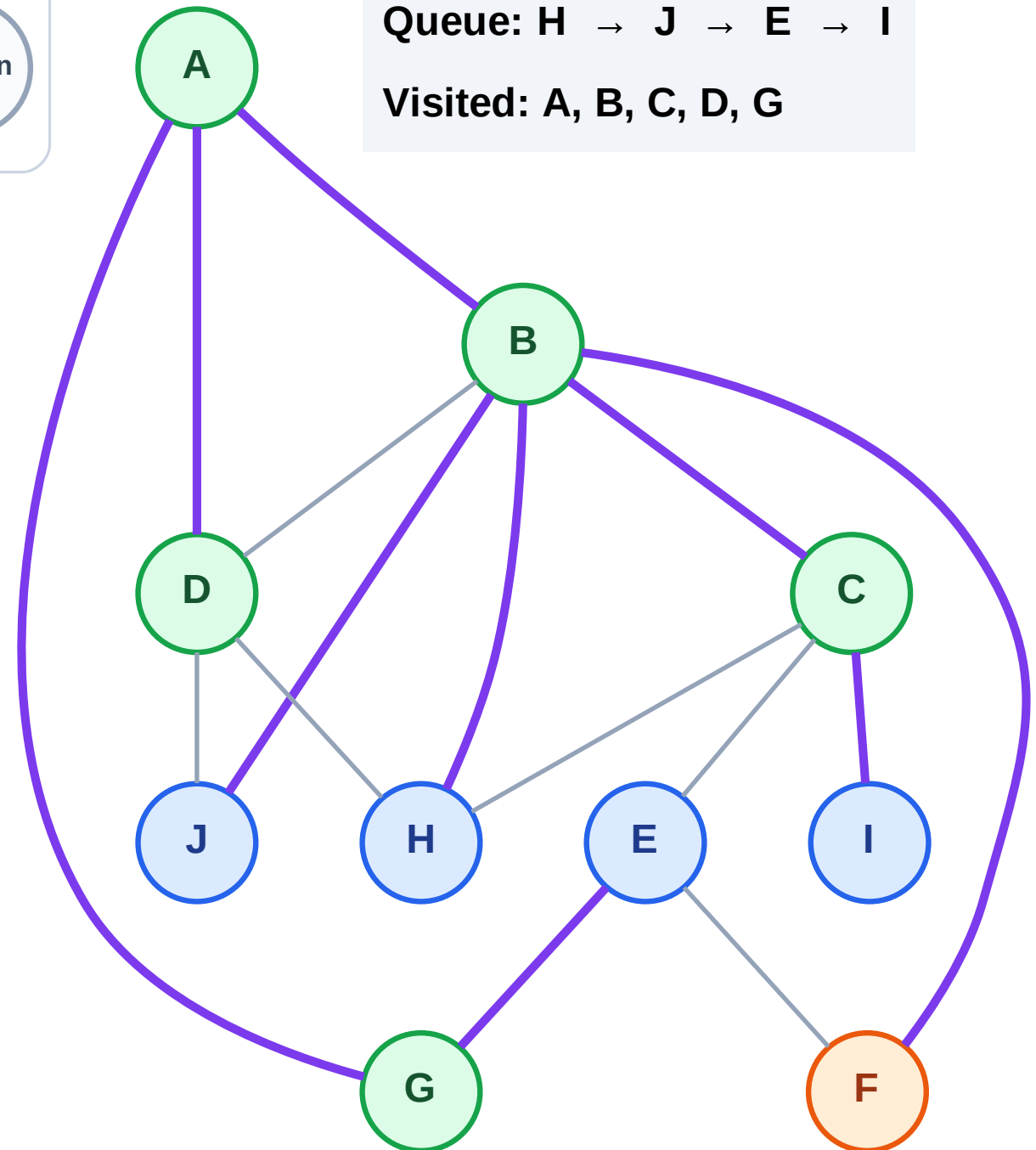
Step 12: Process node F

Legend



Queue: H → J → E → I

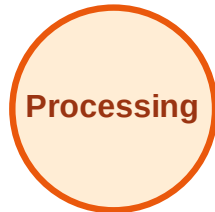
Visited: A, B, C, D, G



BFS Traversal

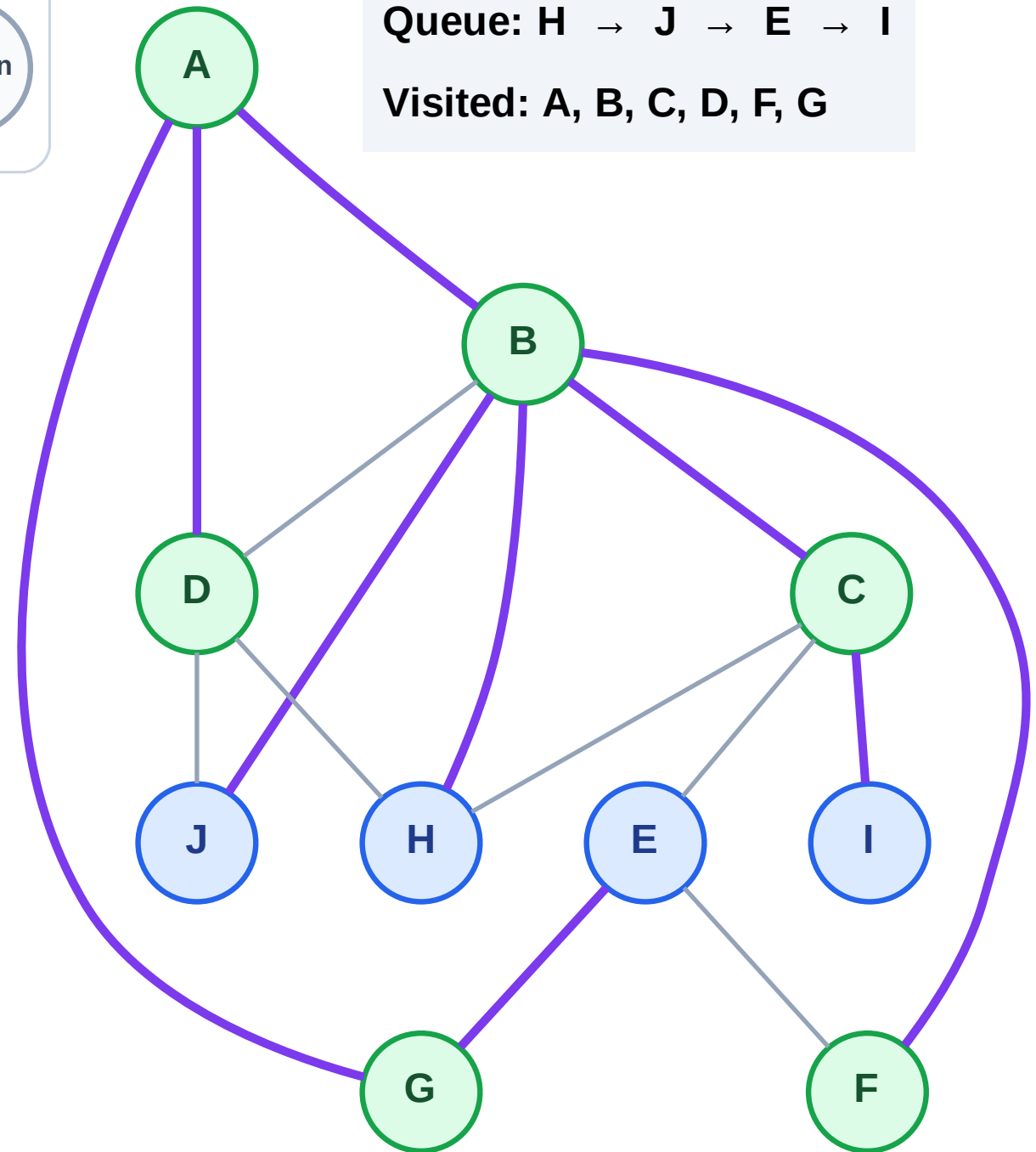
Step 13: No new neighbors from F

Legend



Queue: H → J → E → I

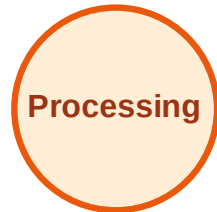
Visited: A, B, C, D, F, G



BFS Traversal

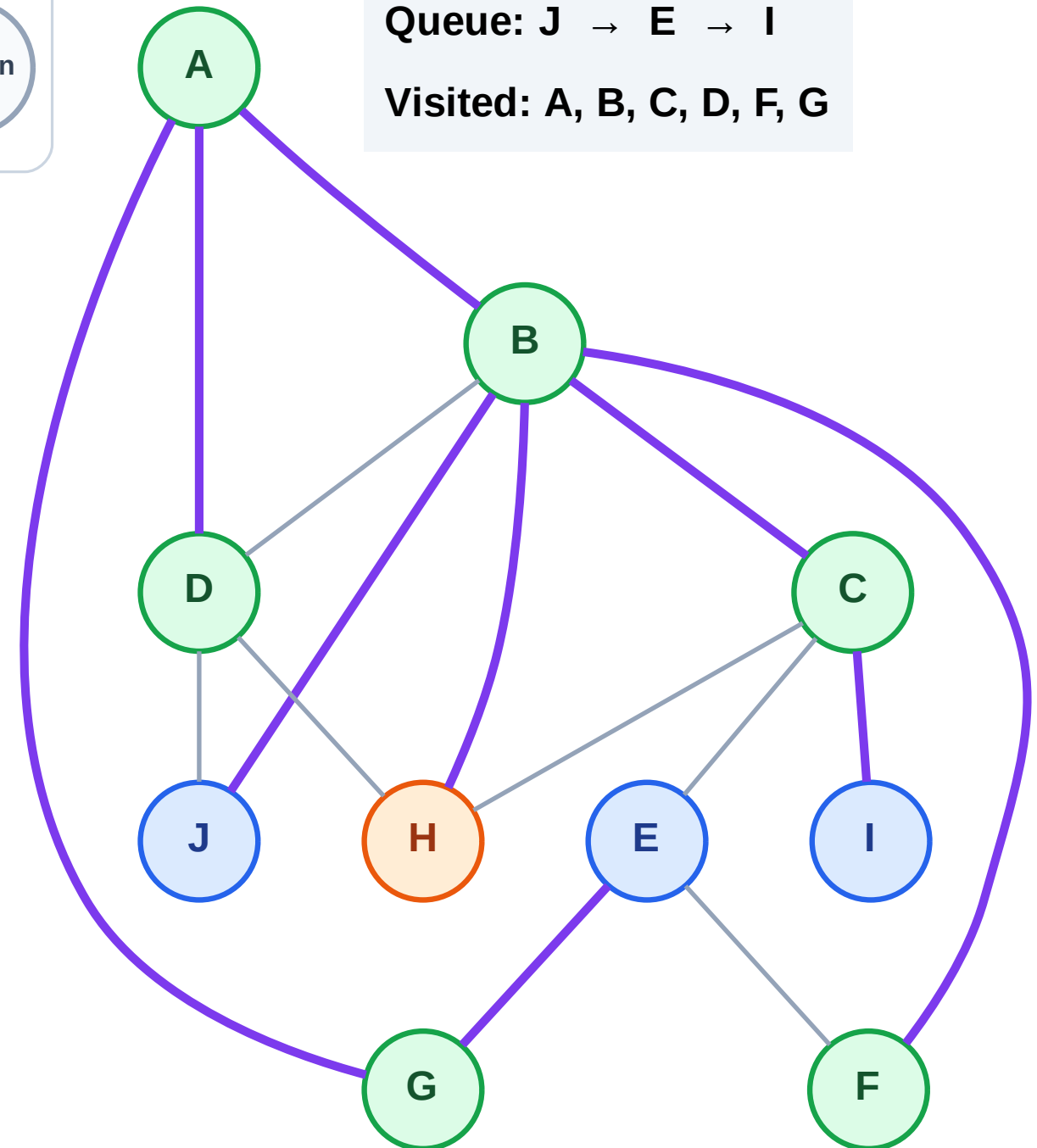
Step 14: Process node H

Legend



Queue: J → E → I

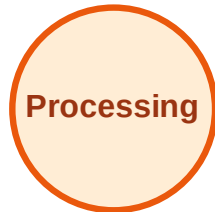
Visited: A, B, C, D, F, G



BFS Traversal

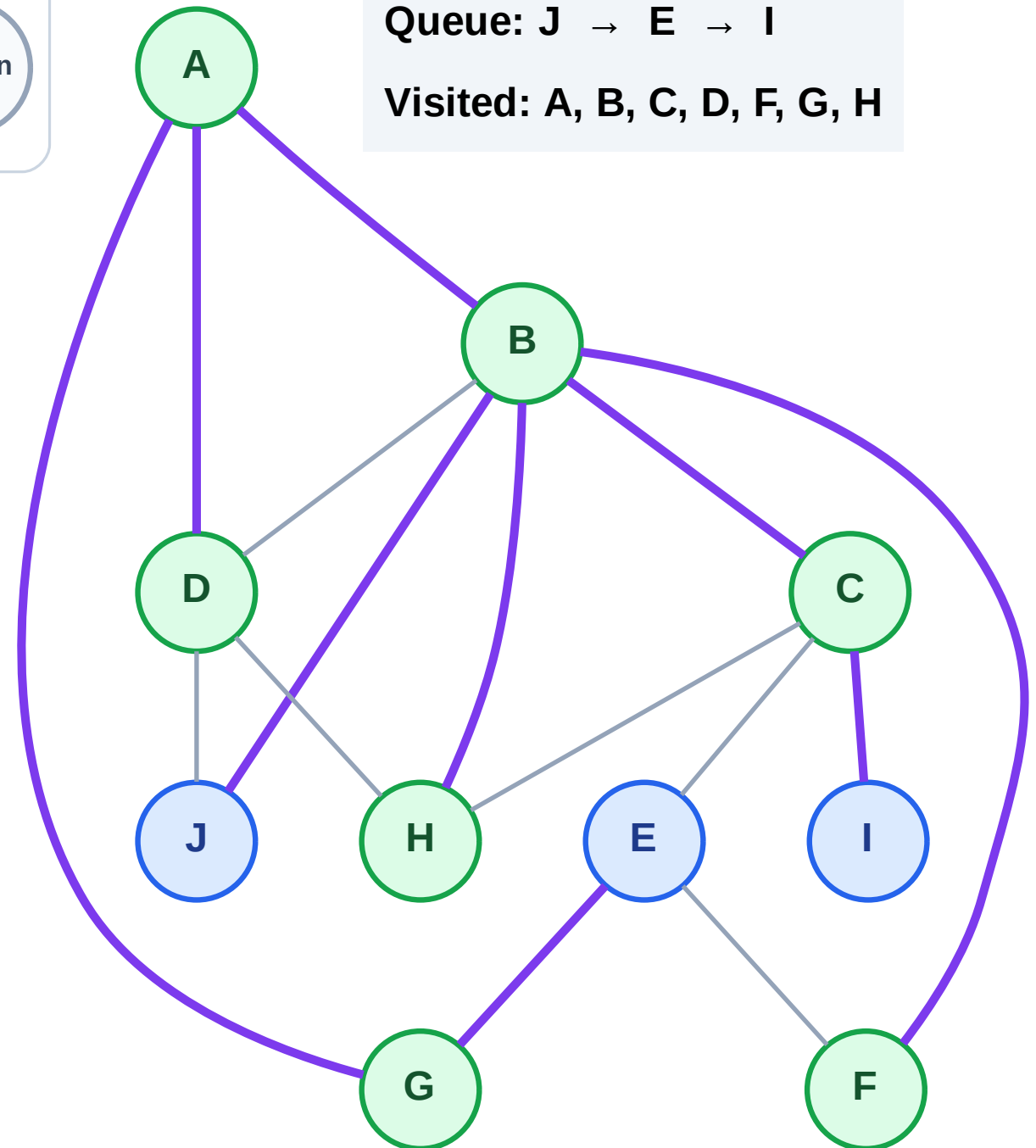
Step 15: No new neighbors from H

Legend



Queue: J → E → I

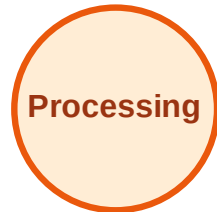
Visited: A, B, C, D, F, G, H



BFS Traversal

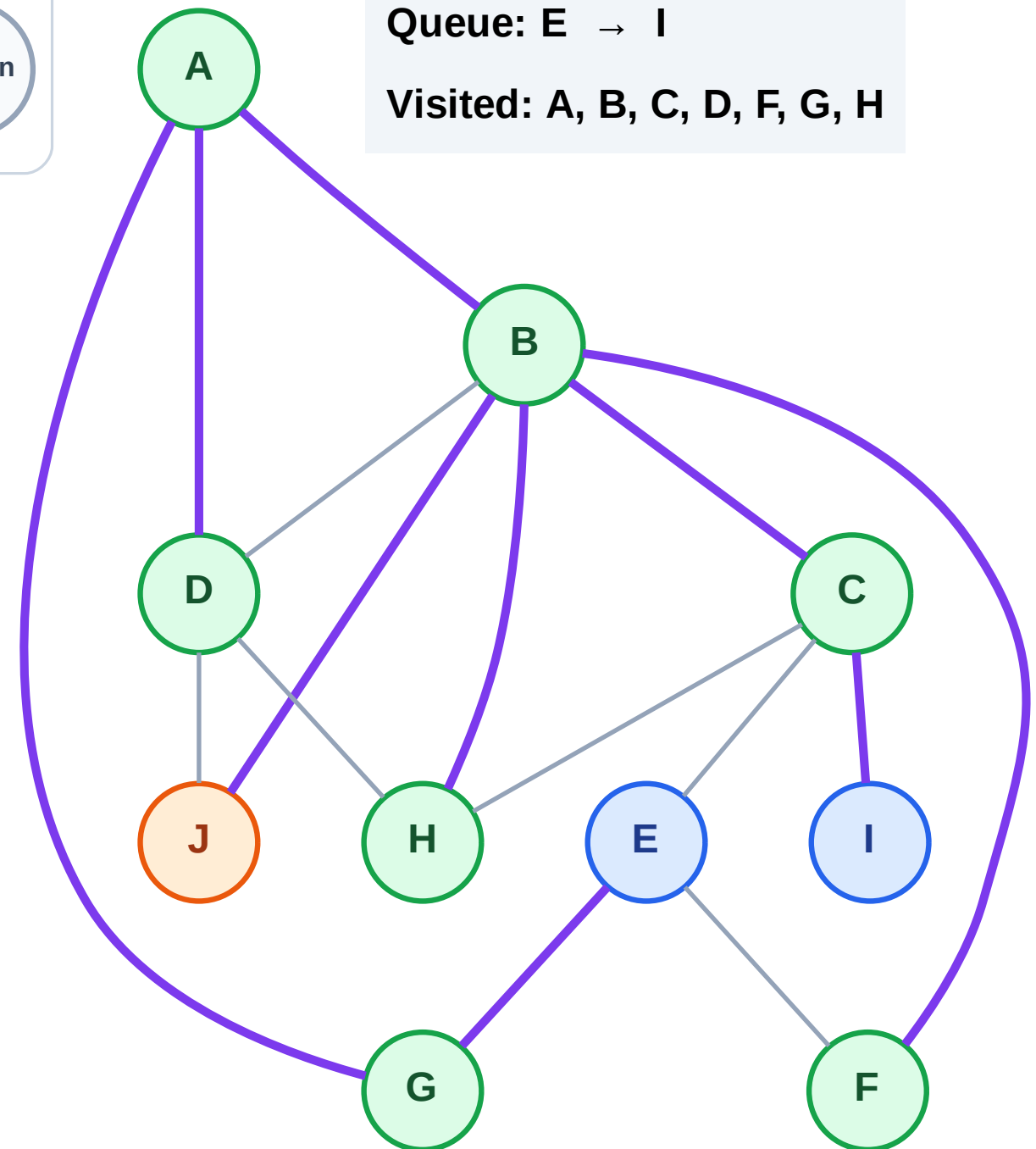
Step 16: Process node J

Legend



Queue: E → I

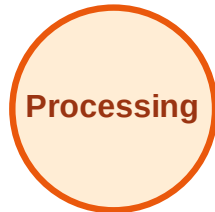
Visited: A, B, C, D, F, G, H



BFS Traversal

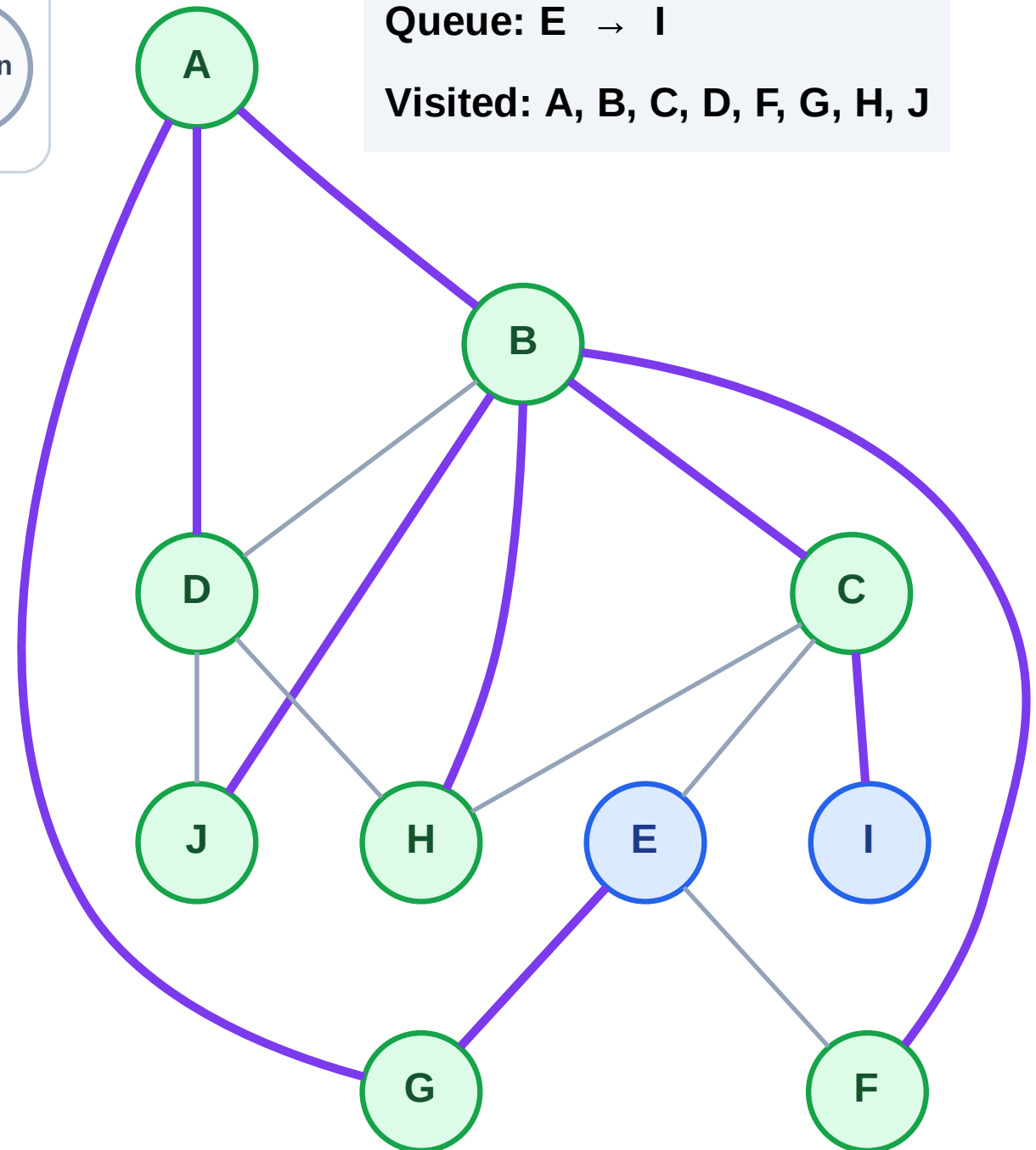
Step 17: No new neighbors from J

Legend



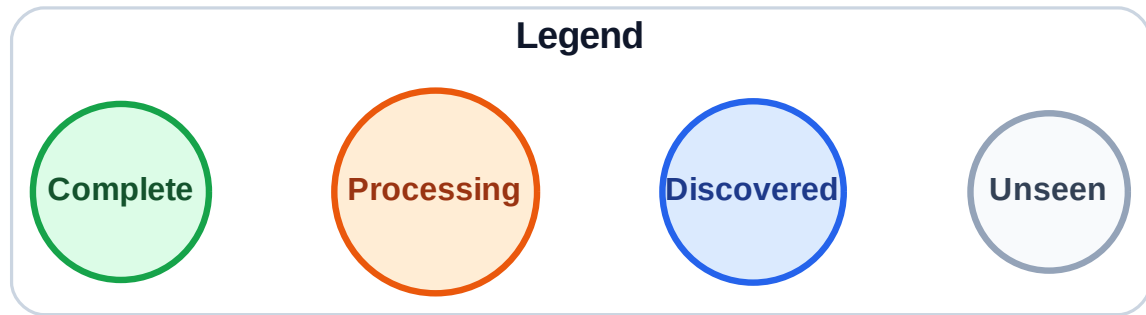
Queue: E → I

Visited: A, B, C, D, F, G, H, J



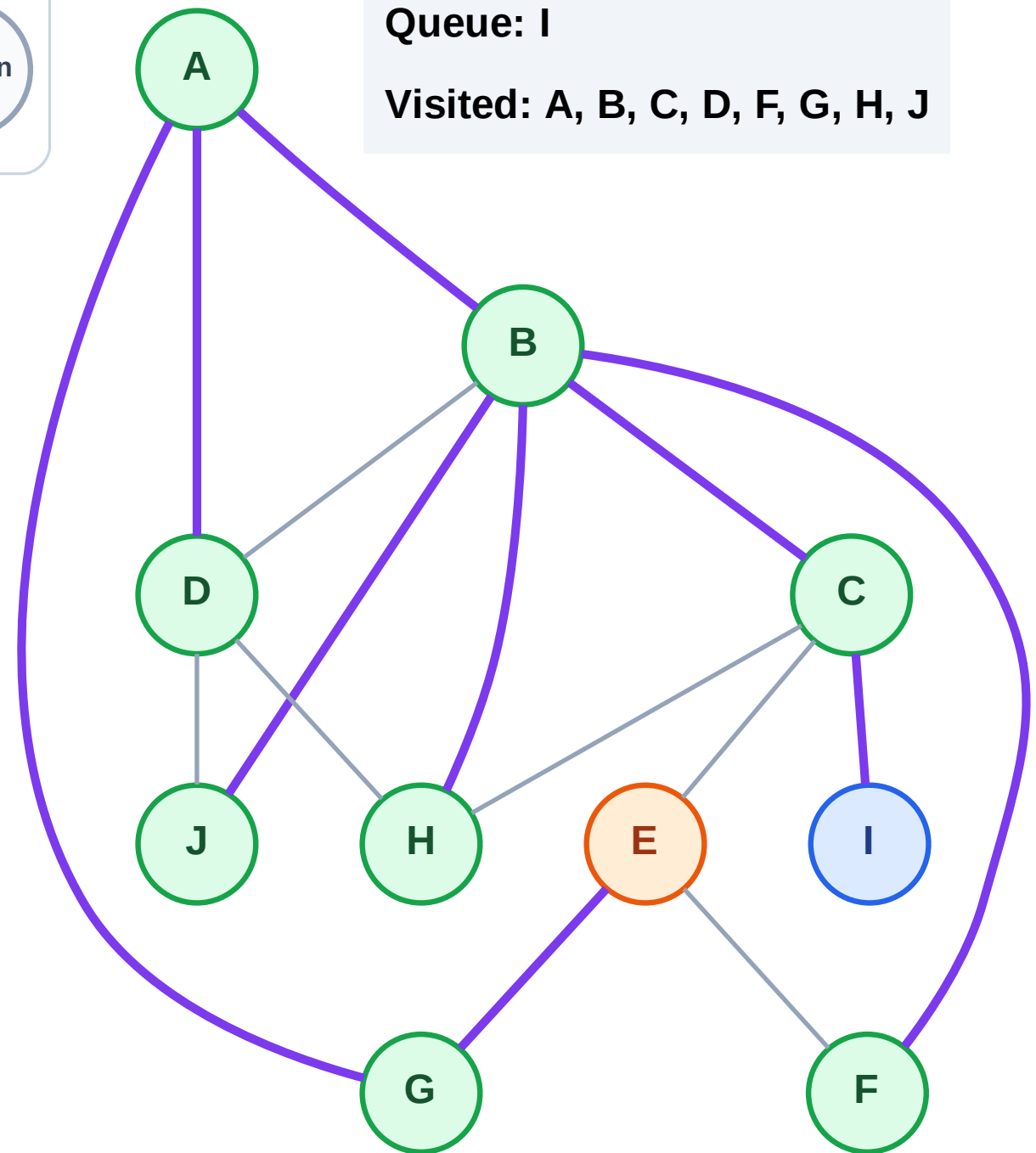
Step 18: Process node E

Step 18: Process node E



Queue: I

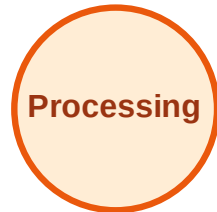
Visited: A, B, C, D, F, G, H, J



BFS Traversal

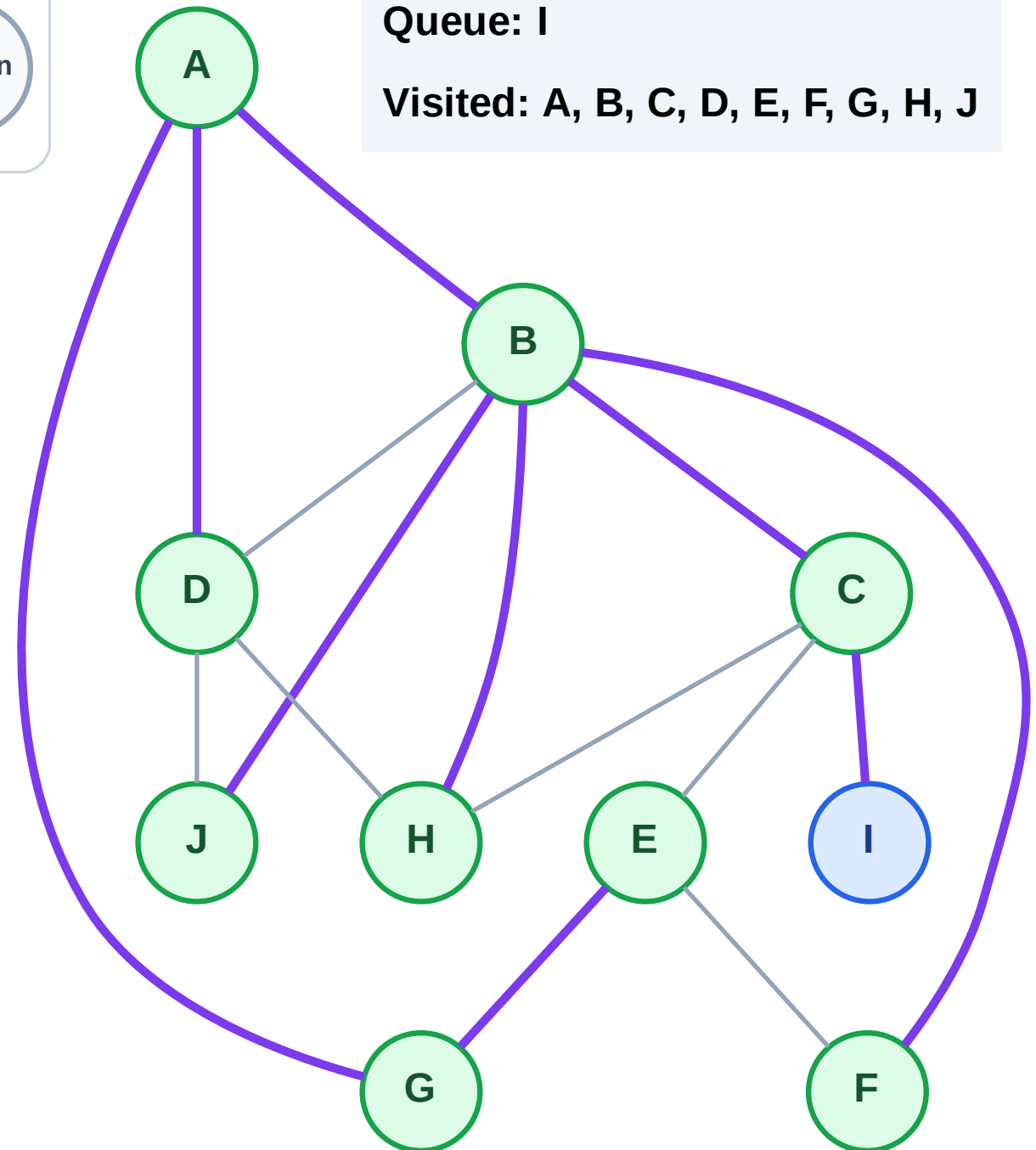
Step 19: No new neighbors from E

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



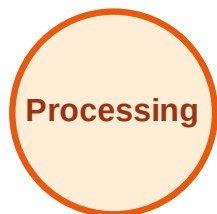
Step 20: Process node I

Step 20: Process node I

Legend



Complete



Processing



Discovered



Unseen

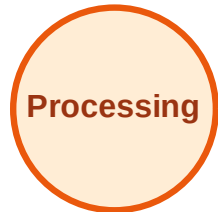
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J

BFS Traversal

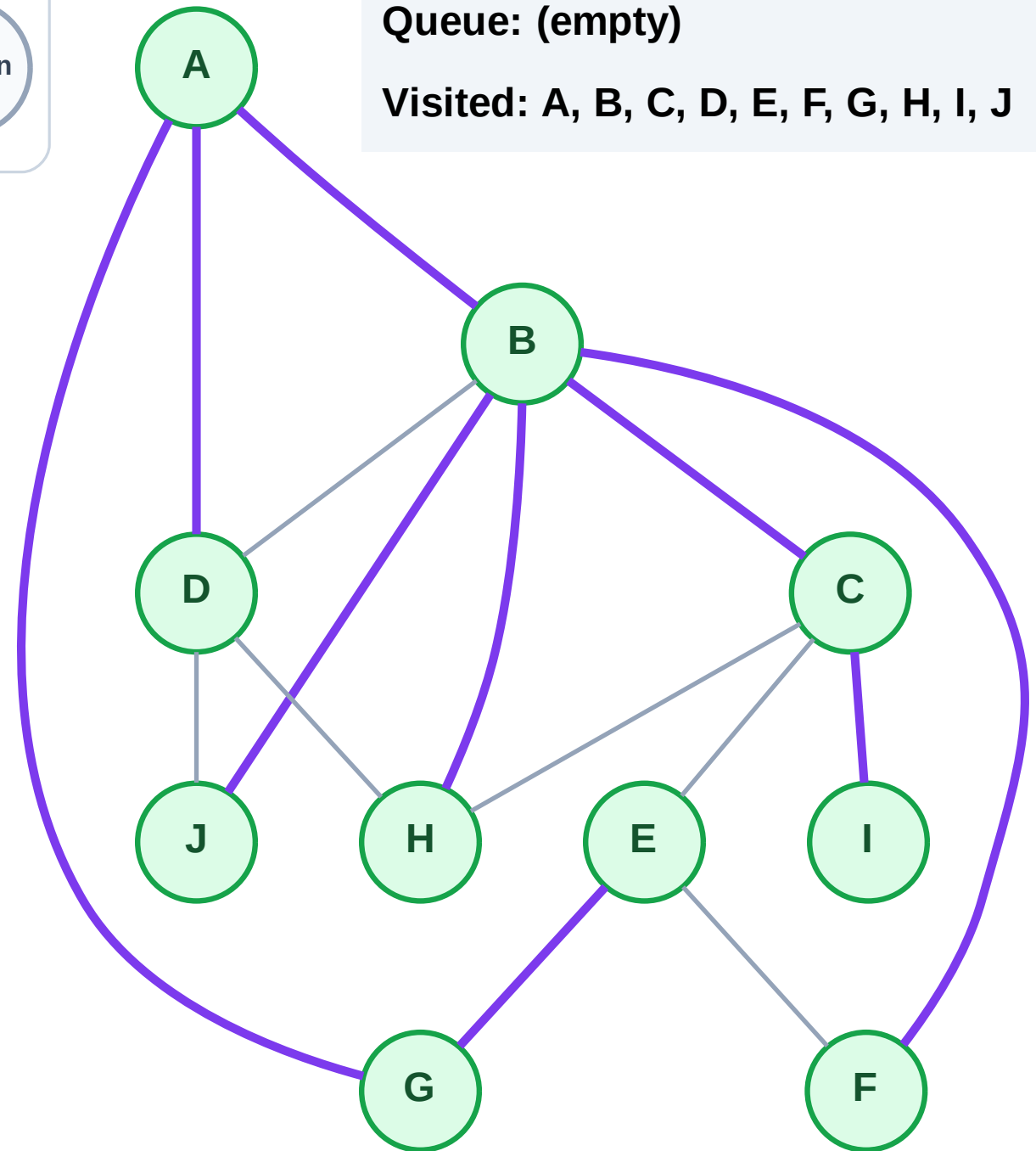
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

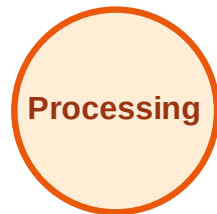


Step 1: Start at A

Step 1: Start at A

Complete

Complete



Processing

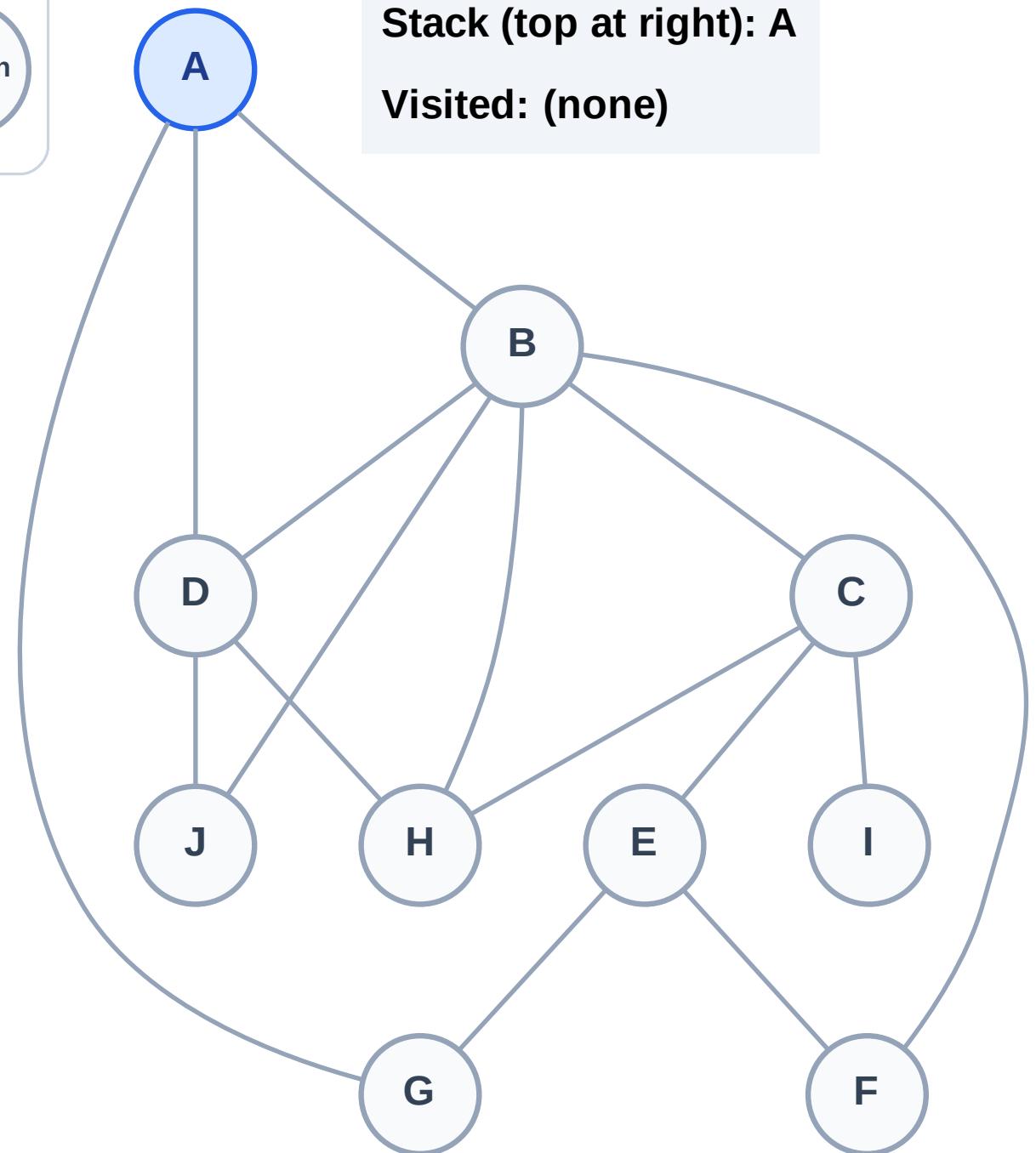


Discovered



Unseen

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

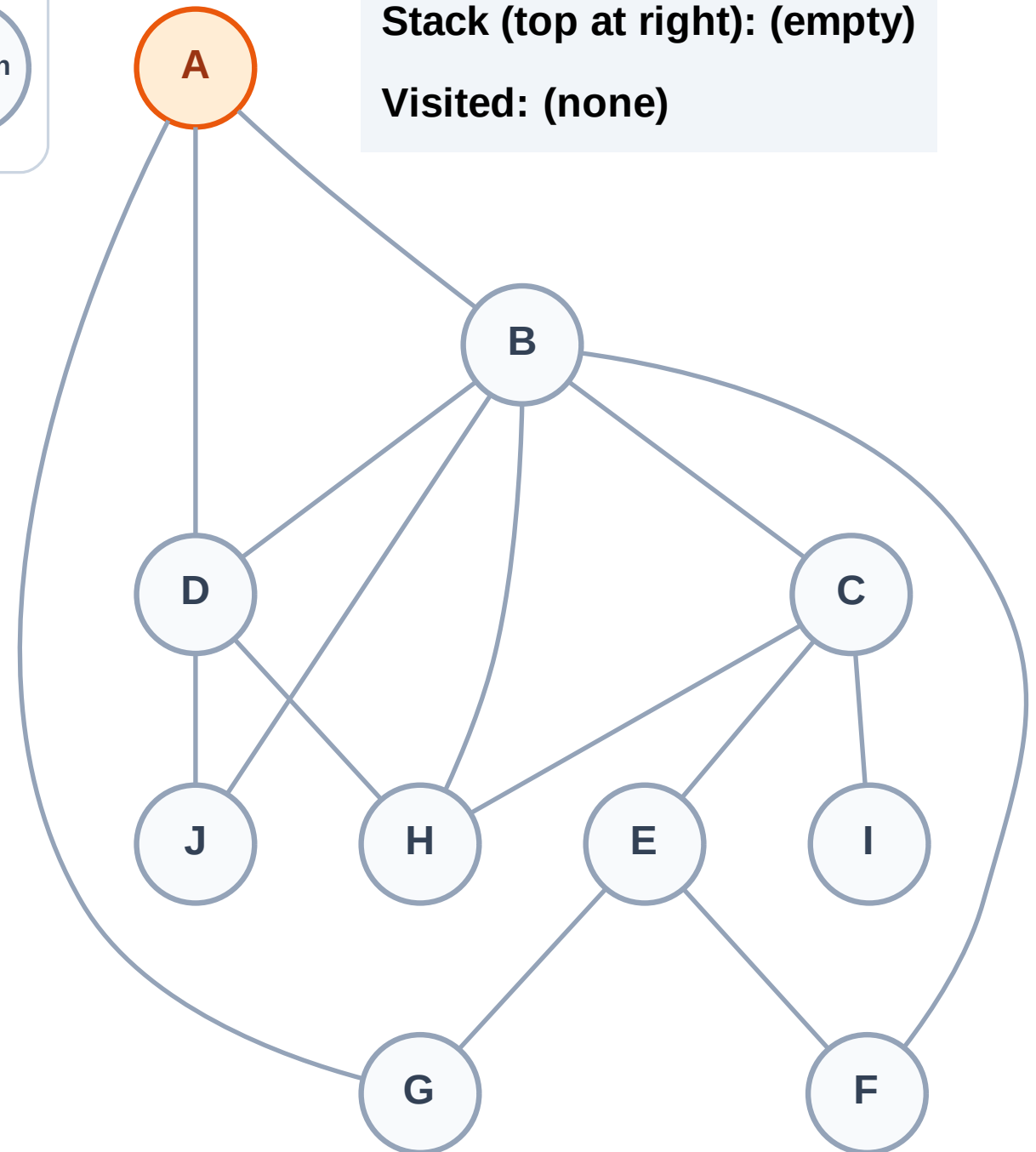
Processing

Discovered

Unseen

Stack (top at right): (empty)

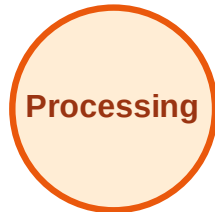
Visited: (none)



DFS Traversal

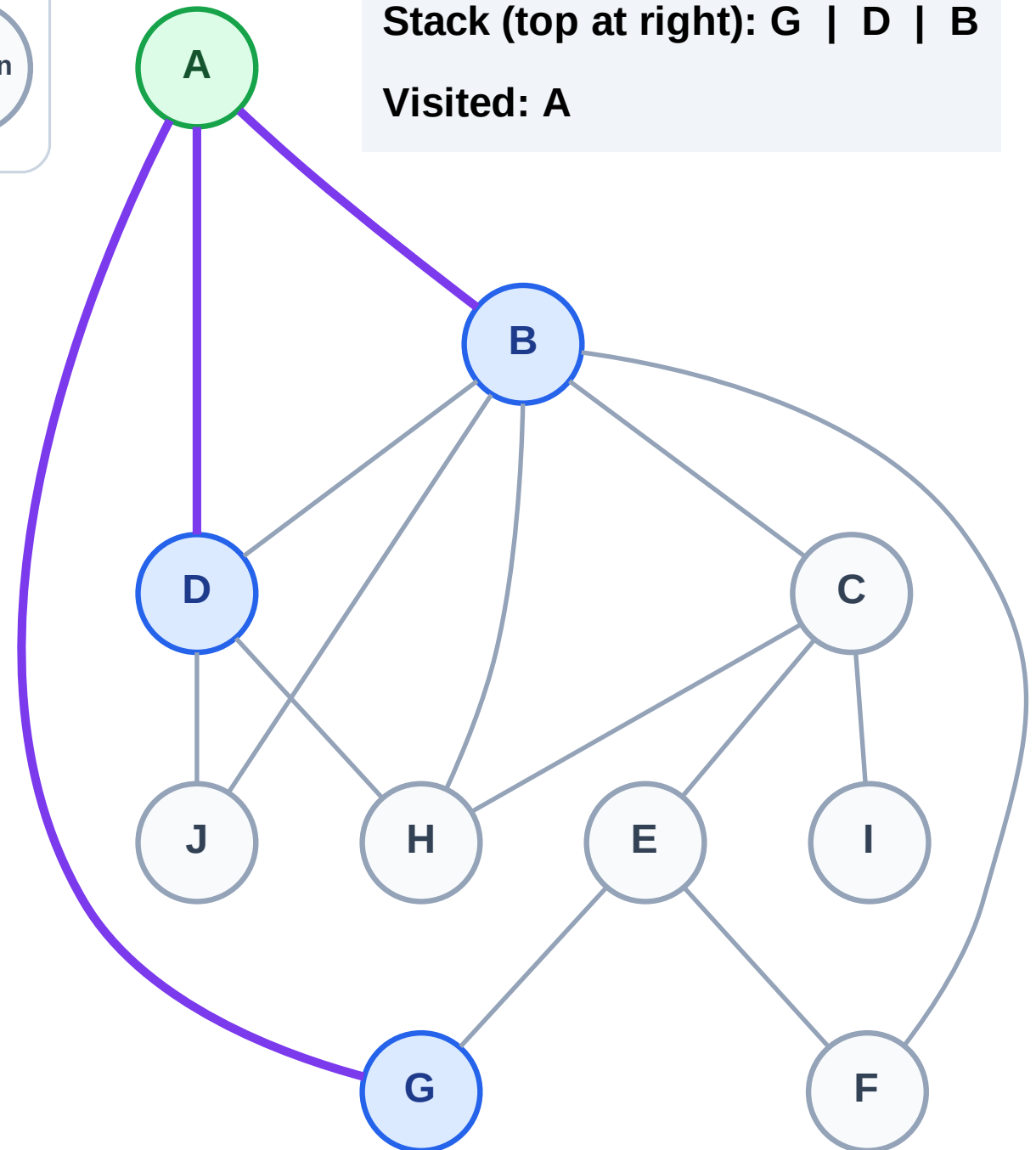
Step 3: Discover G, D, B from A

Legend



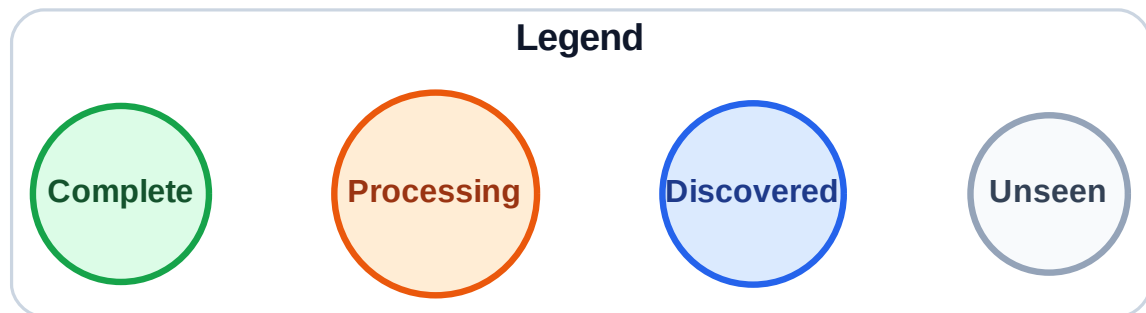
Stack (top at right): G | D | B

Visited: A



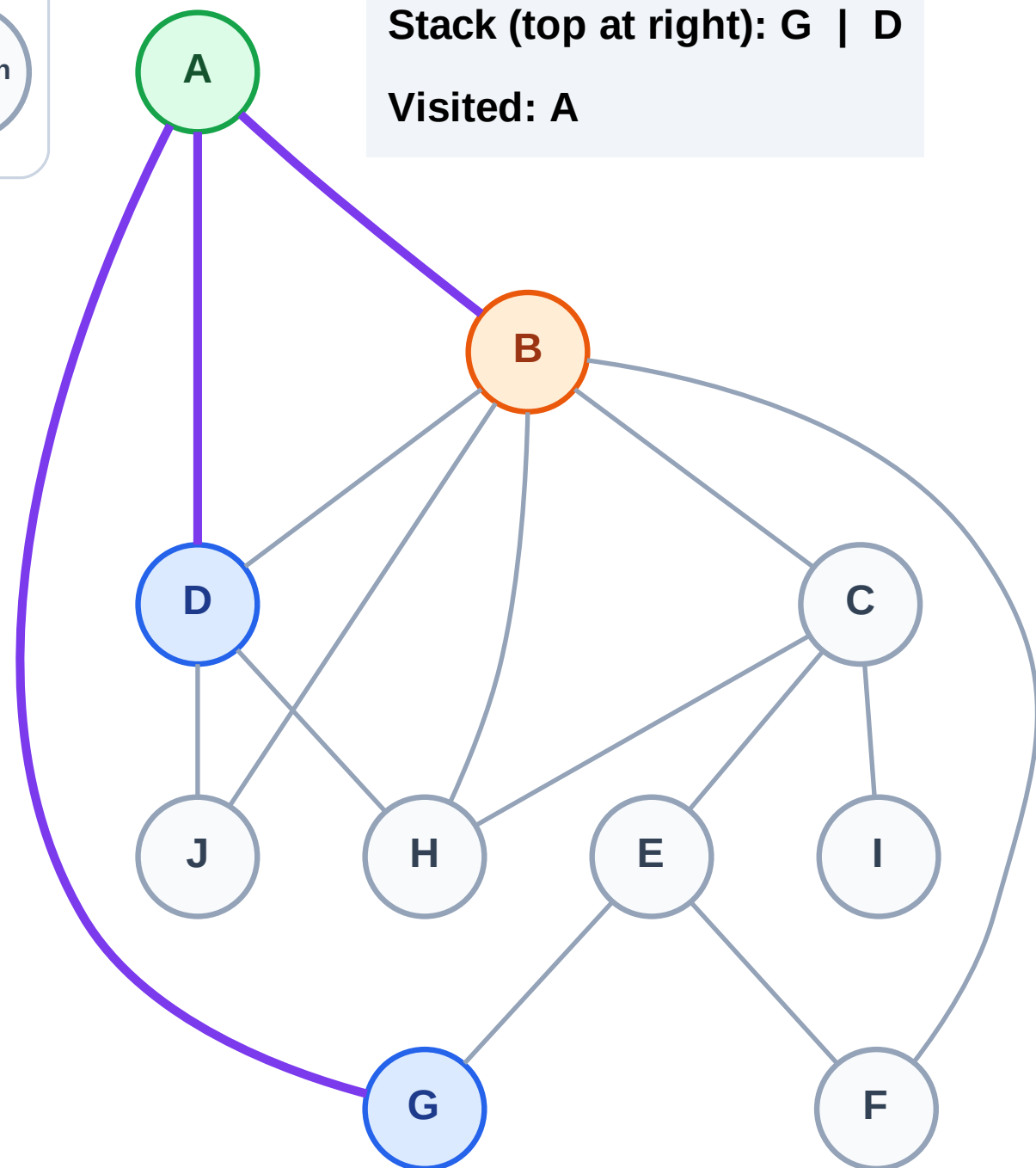
Step 4: Process node B

Step 4: Process node B



Stack (top at right): G | D

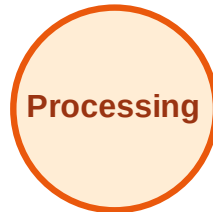
Visited: A



DFS Traversal

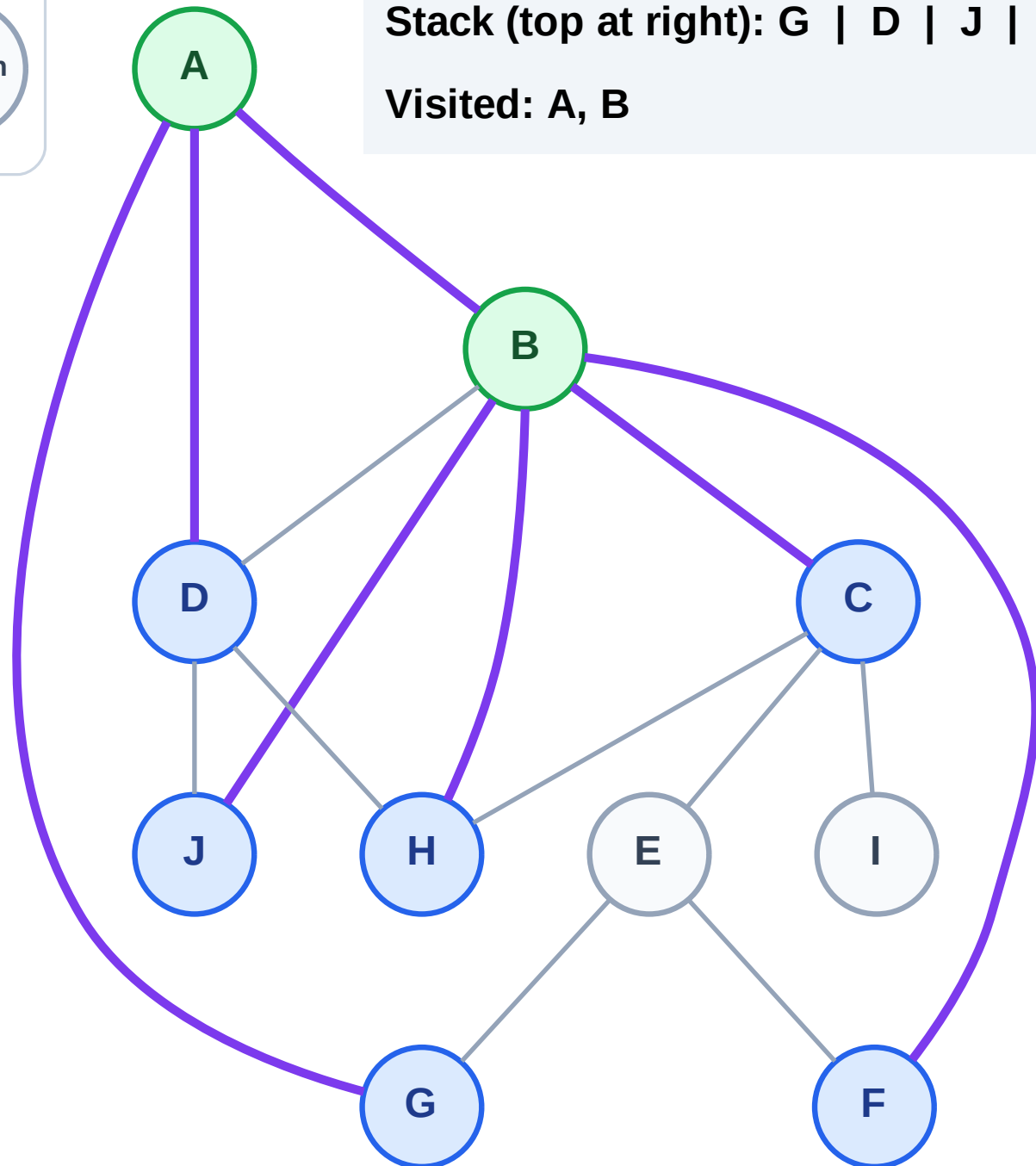
Step 5: Discover J, H, F, C from B

Legend



Stack (top at right): G | D | J | H | F | C

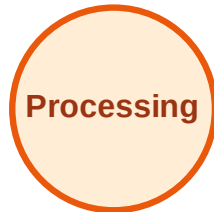
Visited: A, B



DFS Traversal

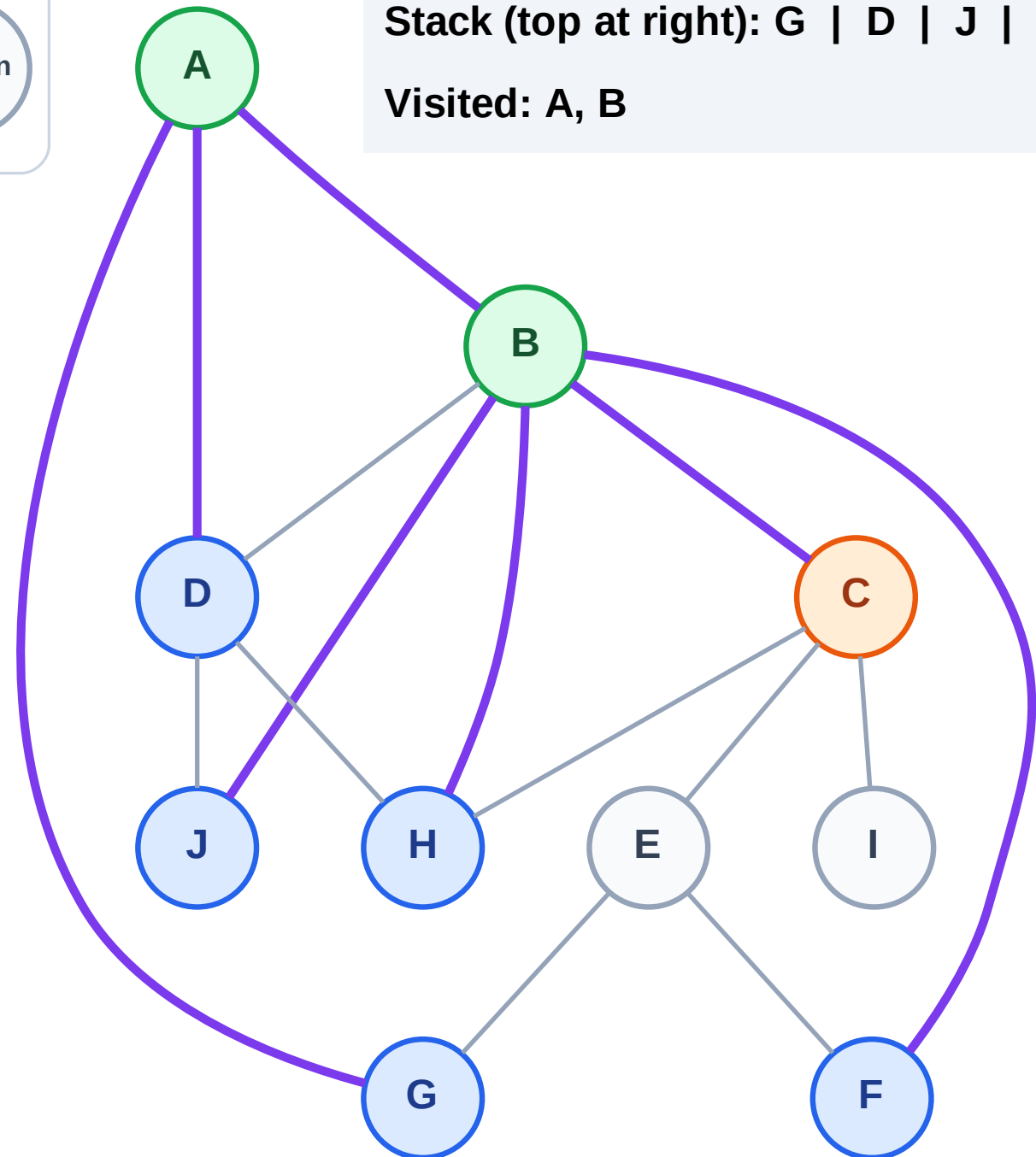
Step 6: Process node C

Legend



Stack (top at right): G | D | J | H | F

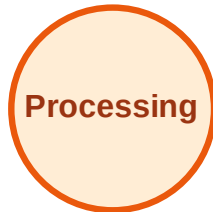
Visited: A, B



DFS Traversal

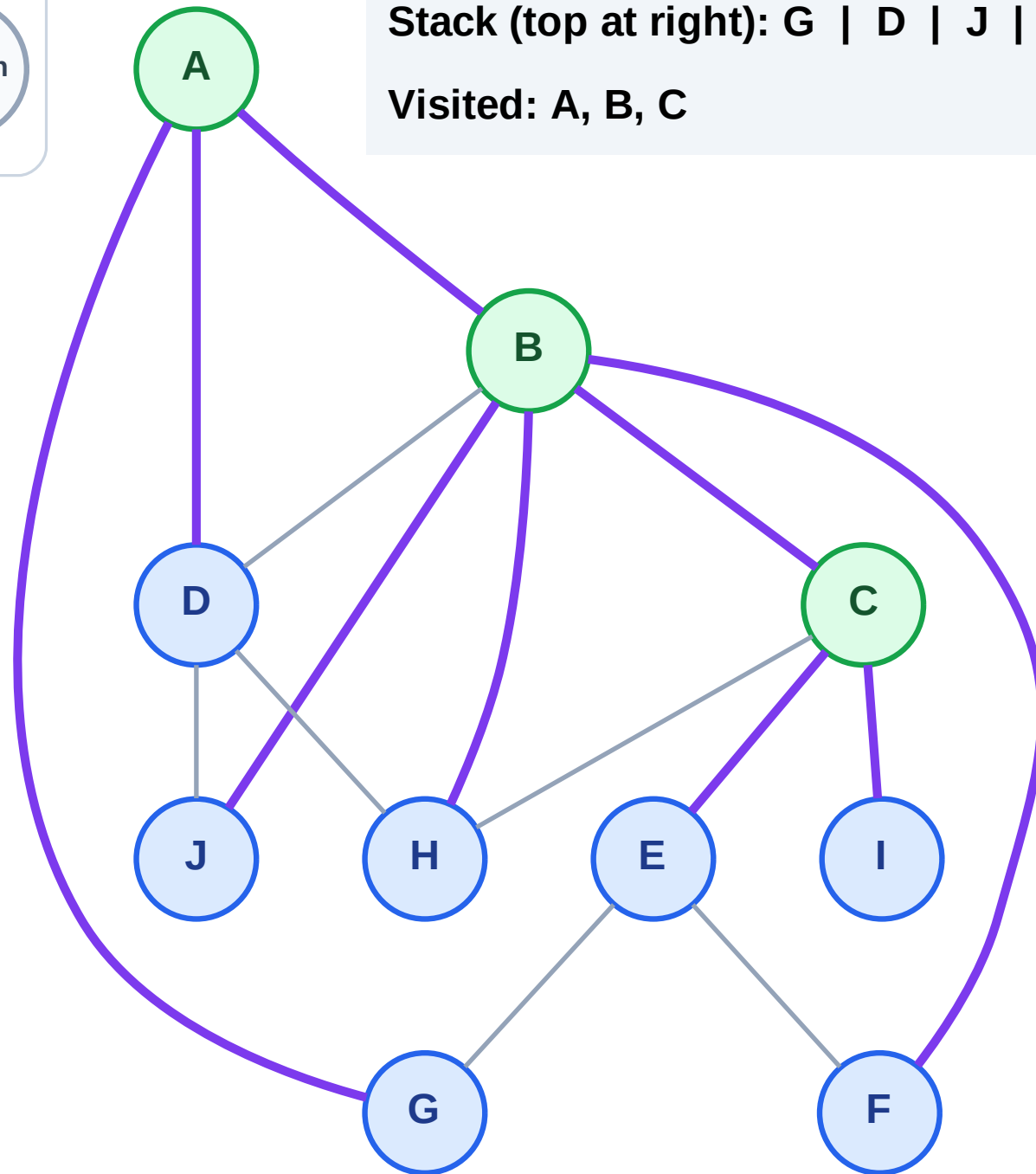
Step 7: Discover I, E from C

Legend

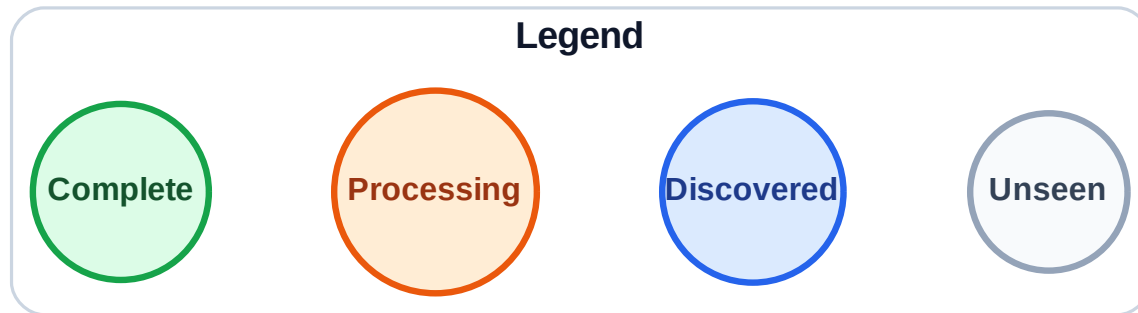


Stack (top at right): G | D | J | H | F | I | E

Visited: A, B, C

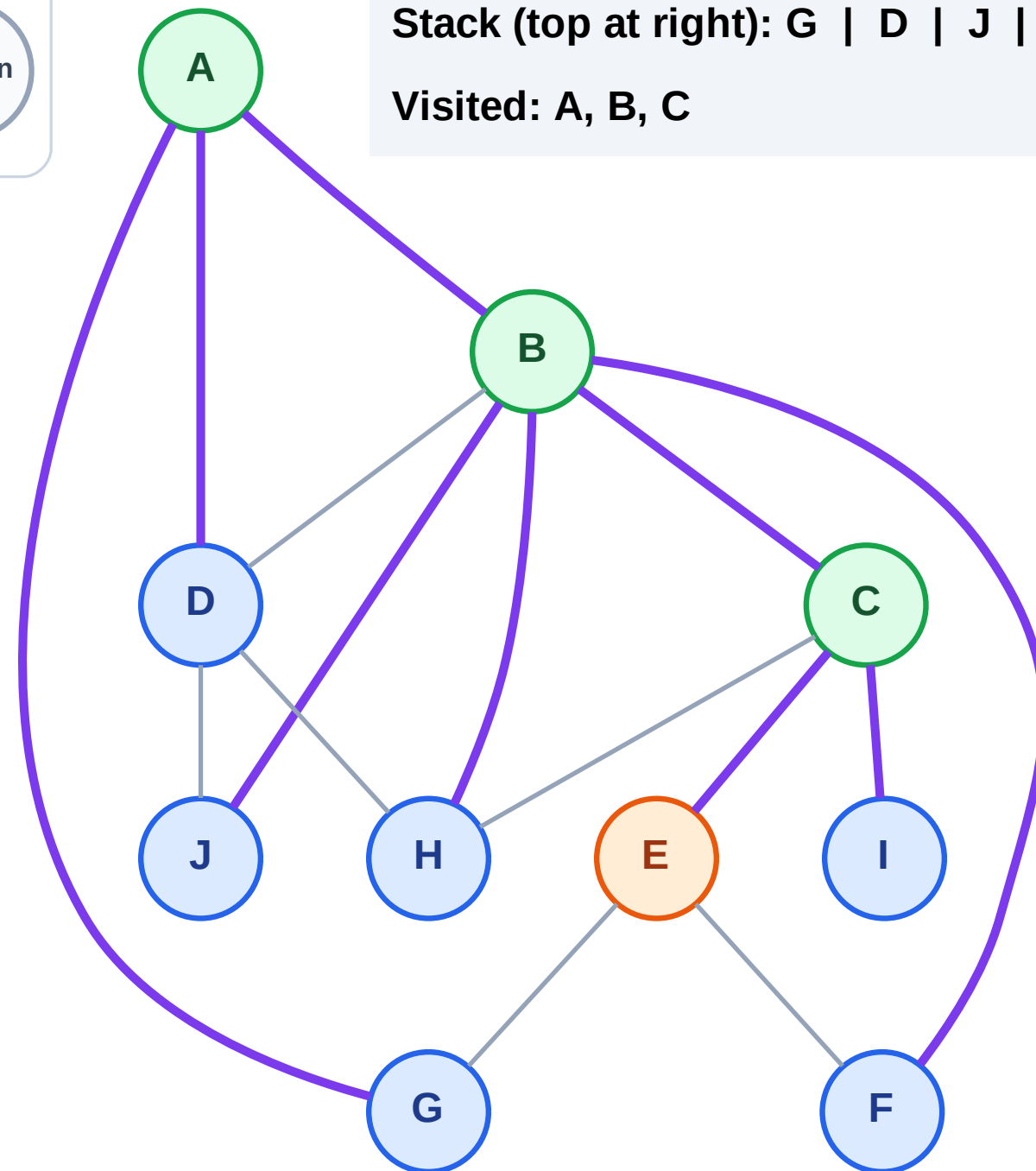


Step 8: Process node E



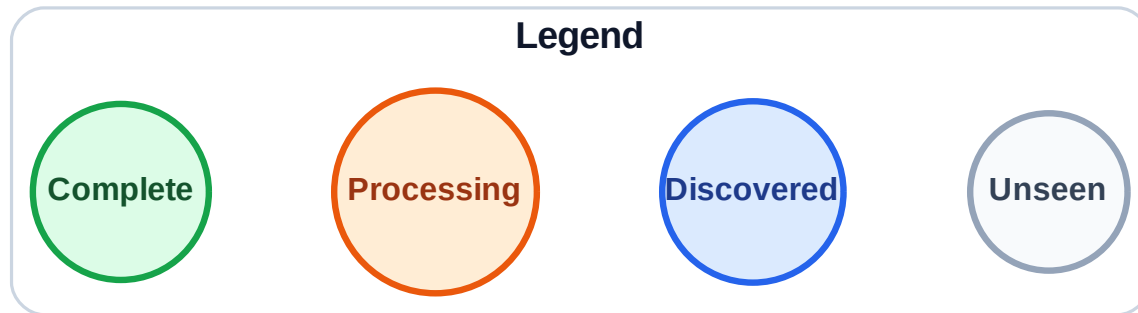
Stack (top at right): G | D | J | H | F | I

Visited: A, B, C



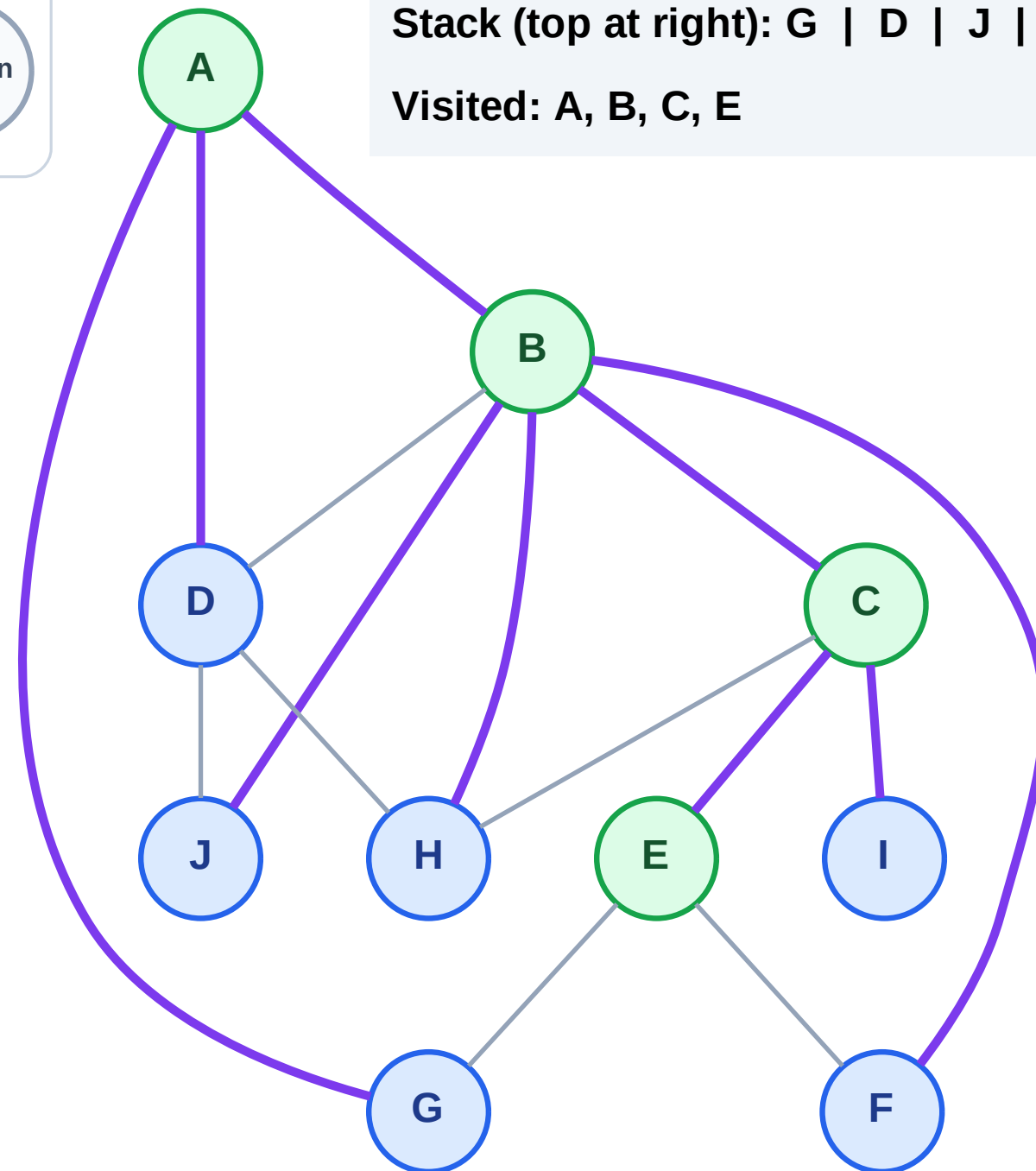
Step 9: No new neighbors from E

Step 9: No new neighbors from E



Stack (top at right): G | D | J | H | F | I

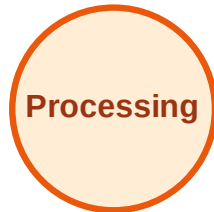
Visited: A, B, C, E



DFS Traversal

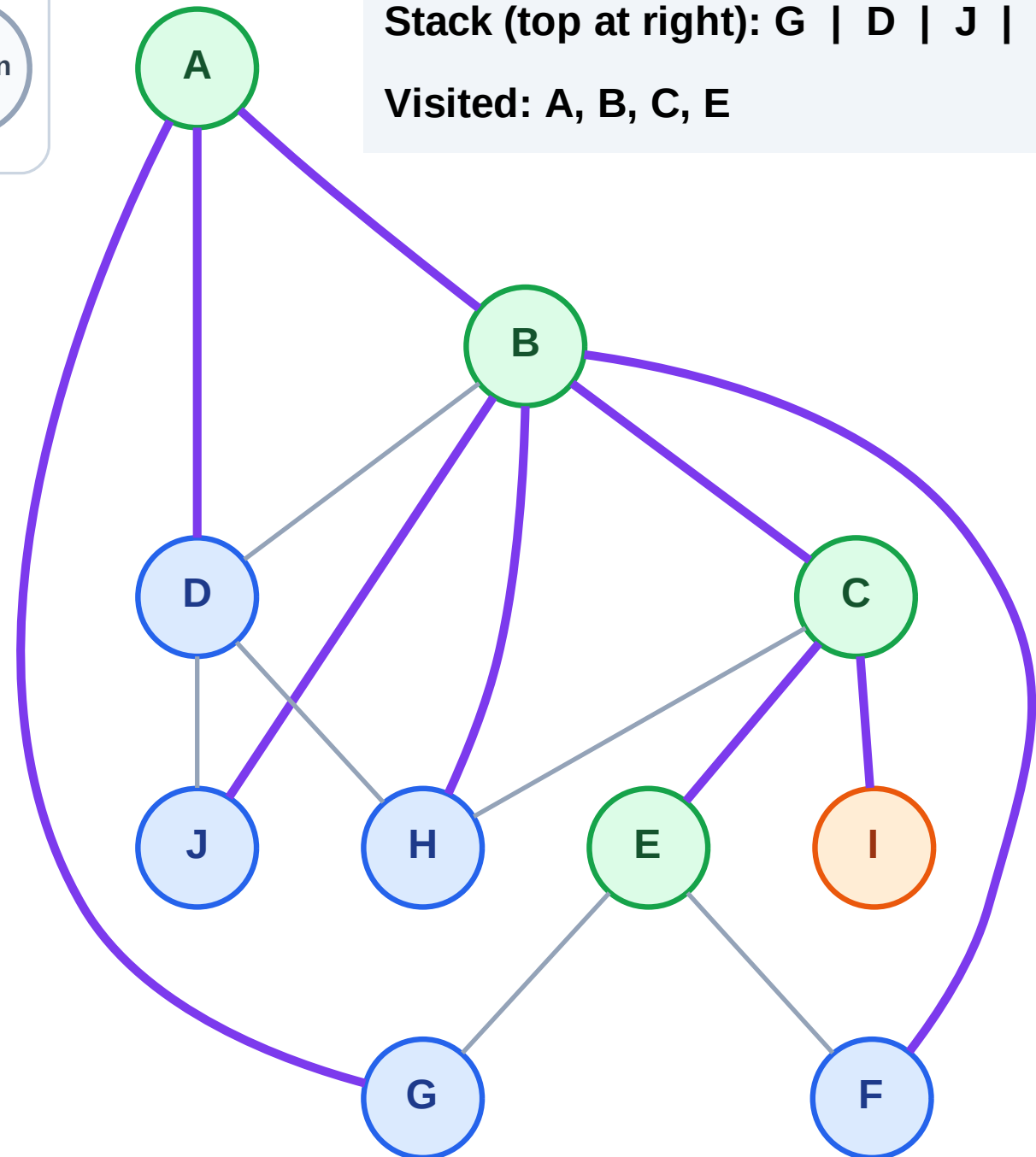
Step 10: Process node I

Legend



Stack (top at right): G | D | J | H | F

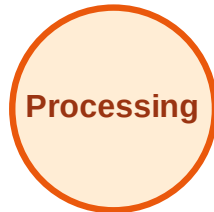
Visited: A, B, C, E



DFS Traversal

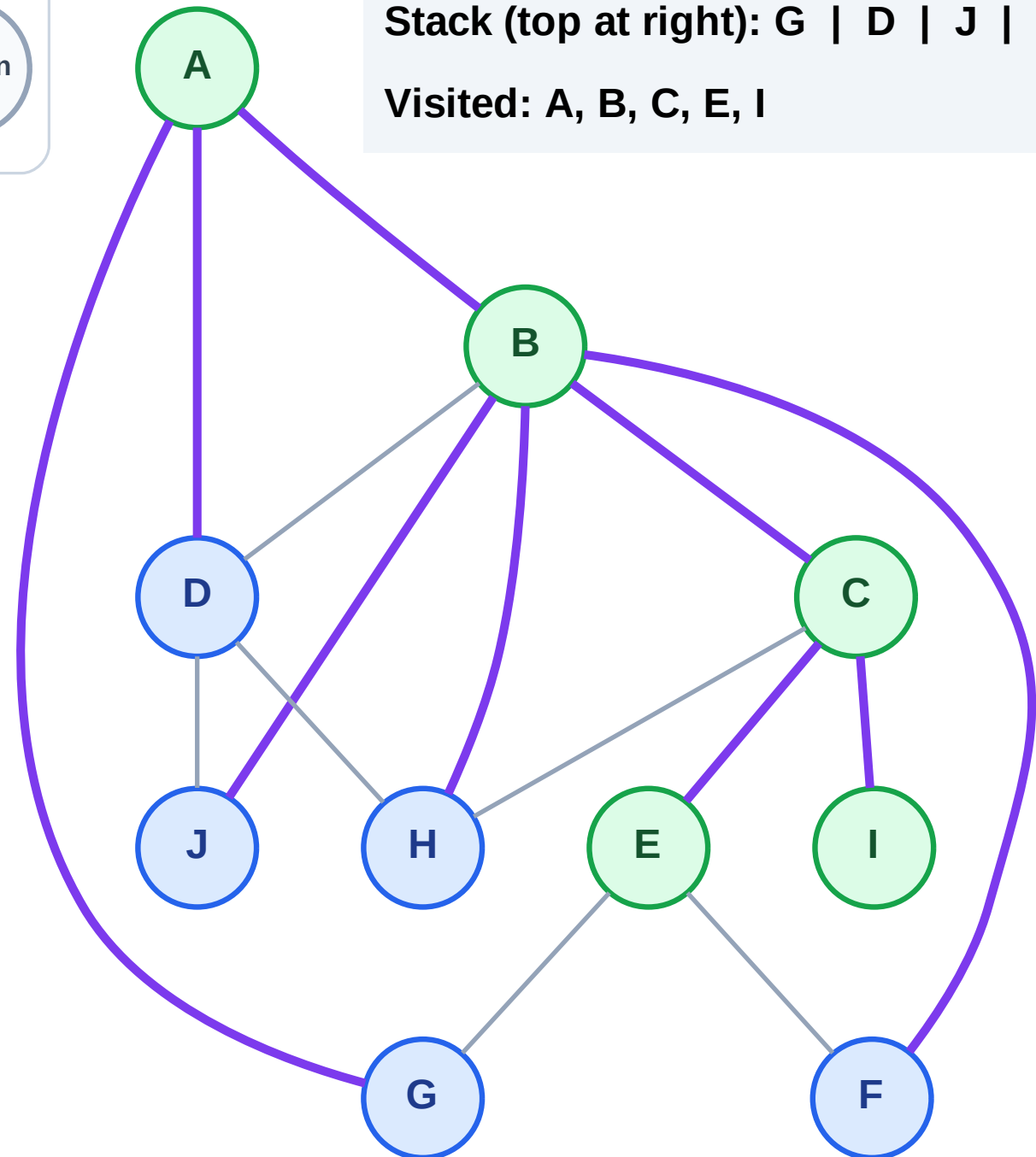
Step 11: No new neighbors from I

Legend



Stack (top at right): G | D | J | H | F

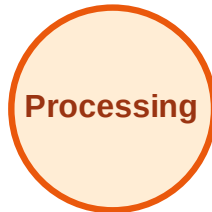
Visited: A, B, C, E, I



DFS Traversal

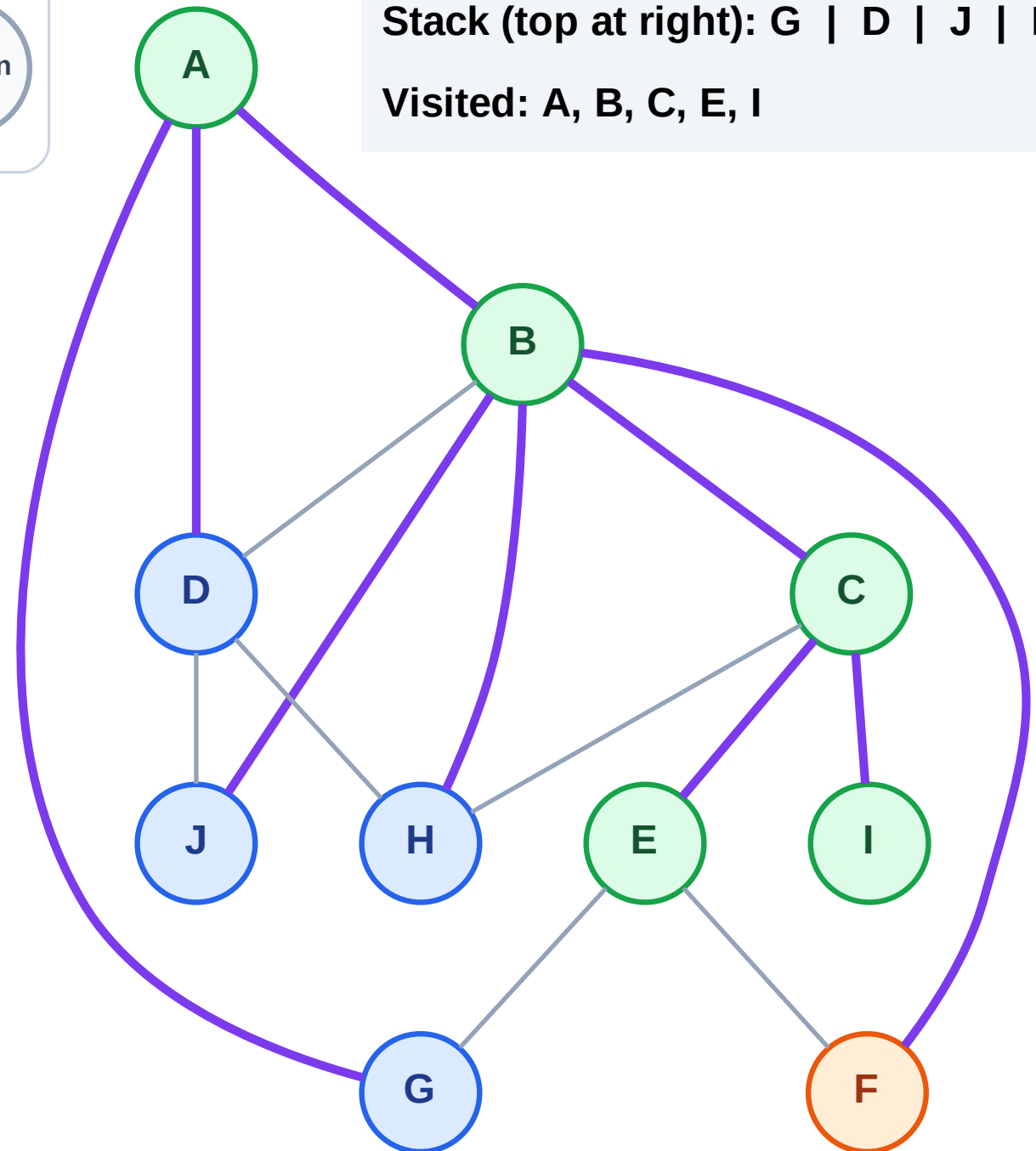
Step 12: Process node F

Legend



Stack (top at right): G | D | J | H

Visited: A, B, C, E, I



DFS Traversal

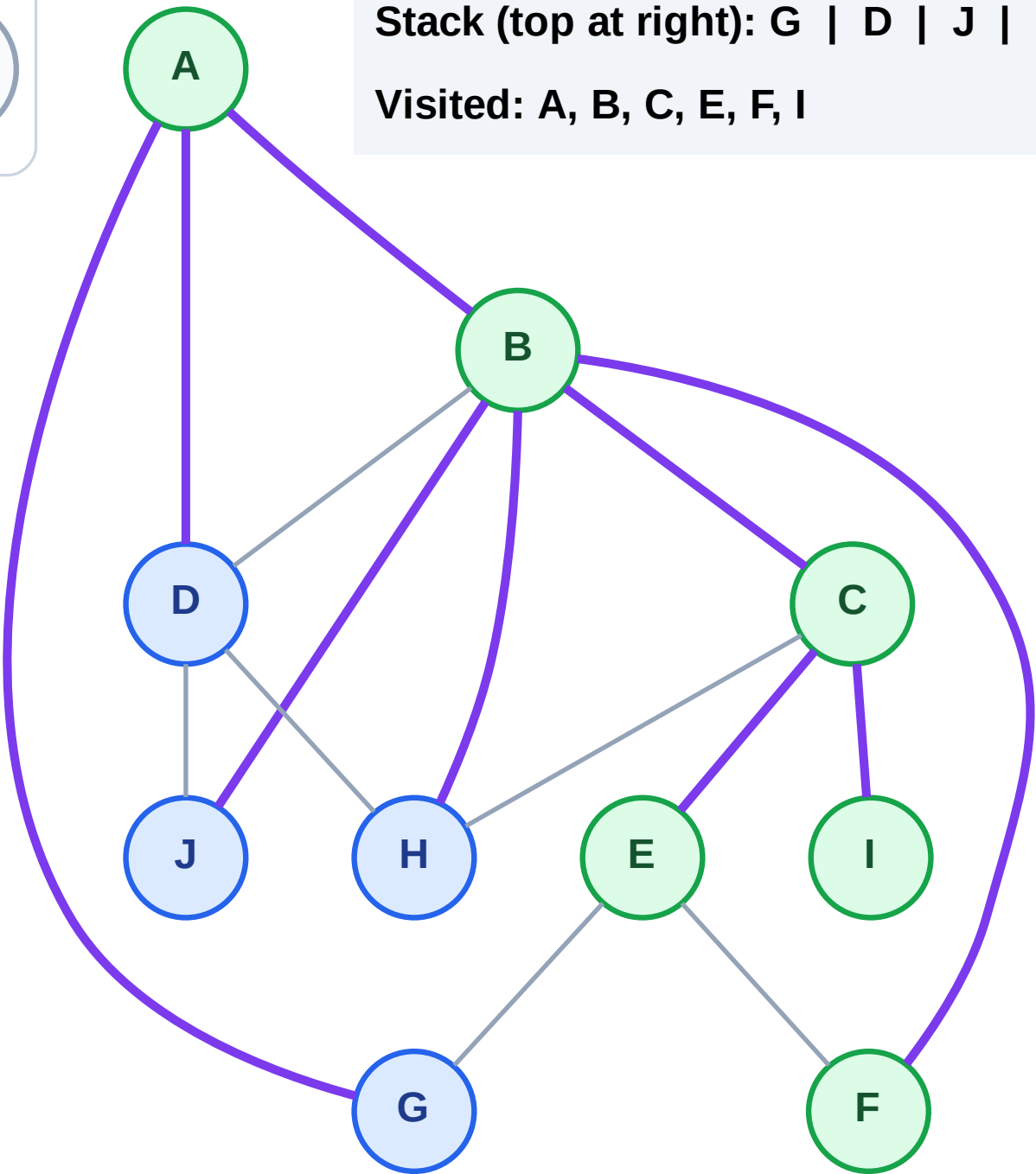
Step 13: No new neighbors from F

Legend



Stack (top at right): G | D | J | H

Visited: A, B, C, E, F, I



Step 14: Process node H

Step 14: Process node H

Legend

Complete

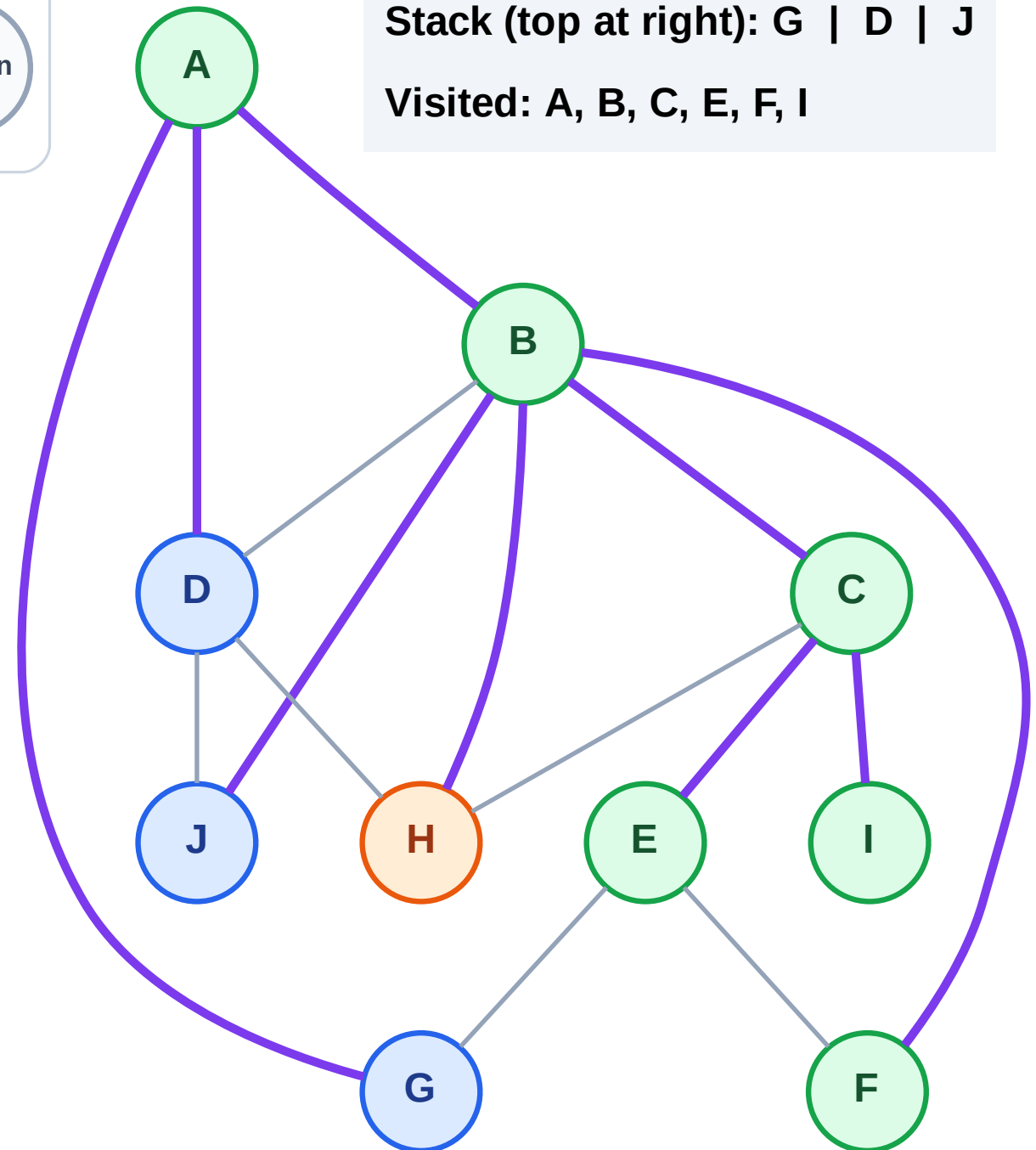
Processing

Discovered

Unseen

Stack (top at right): G | D | J

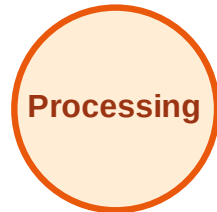
Visited: A, B, C, E, F, I



DFS Traversal

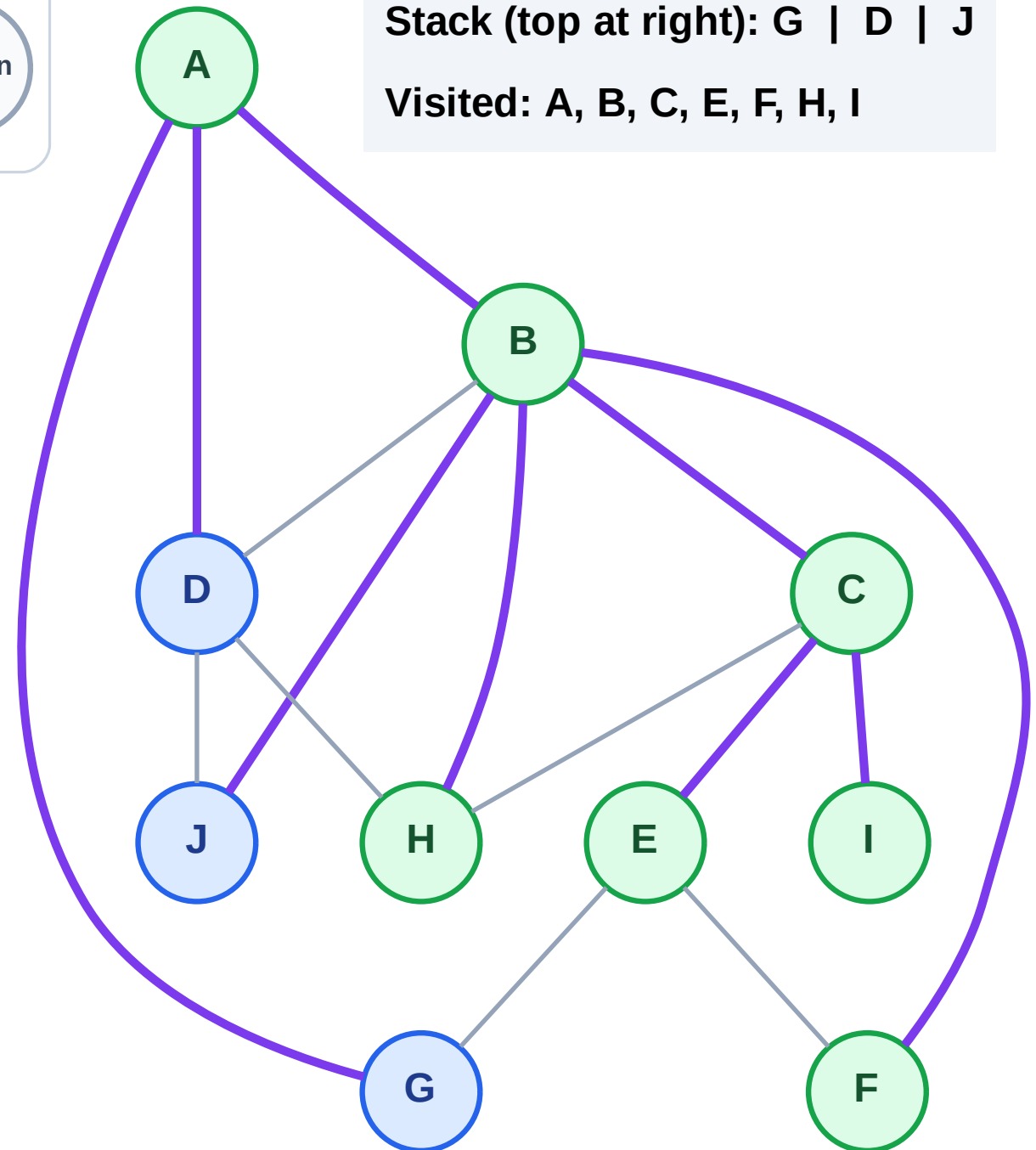
Step 15: No new neighbors from H

Legend



Stack (top at right): G | D | J

Visited: A, B, C, E, F, H, I



DFS Traversal

Step 16: Process node J

Legend

Complete

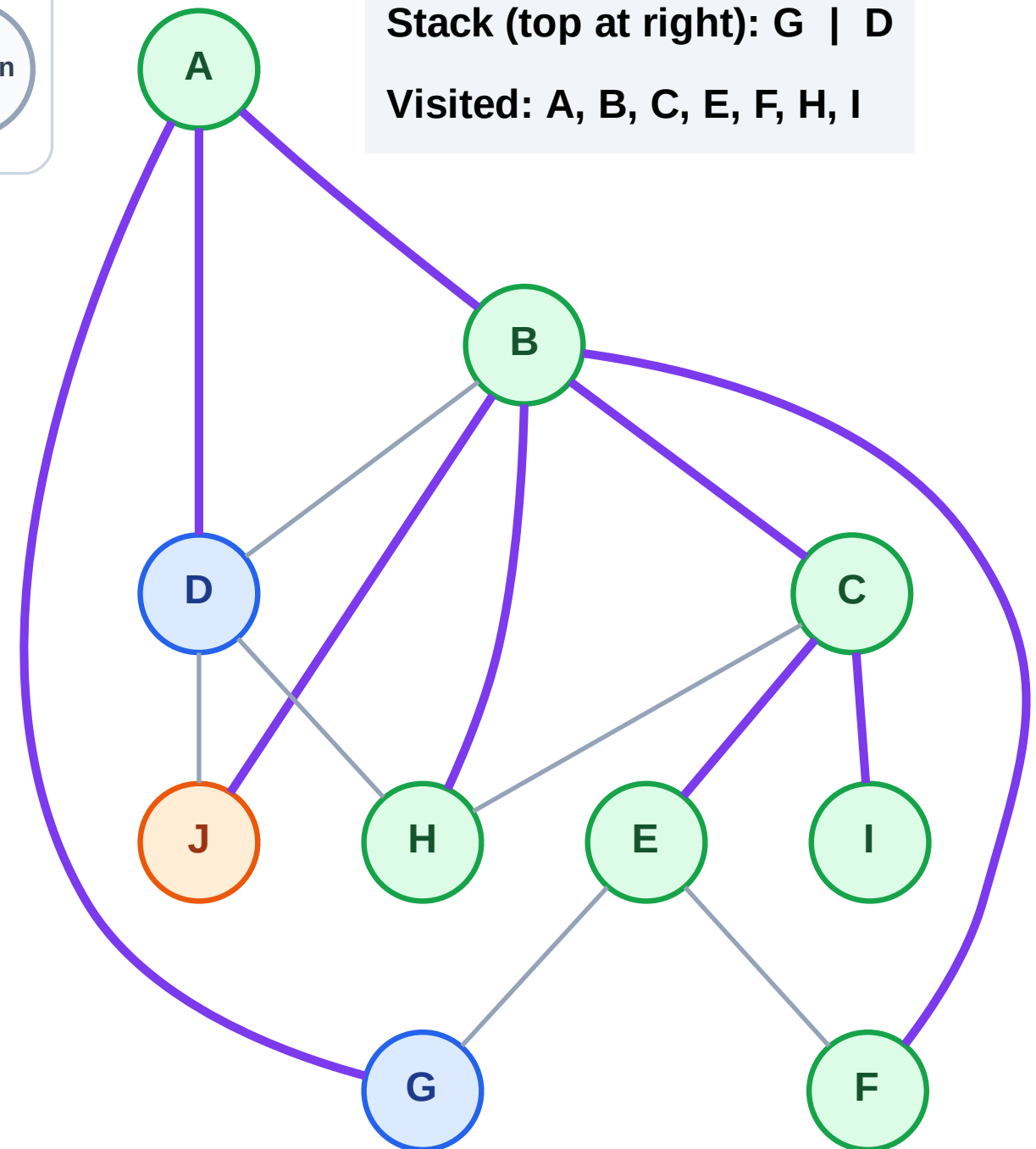
Processing

Discovered

Unseen

Stack (top at right): G | D

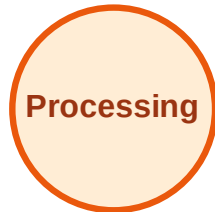
Visited: A, B, C, E, F, H, I



DFS Traversal

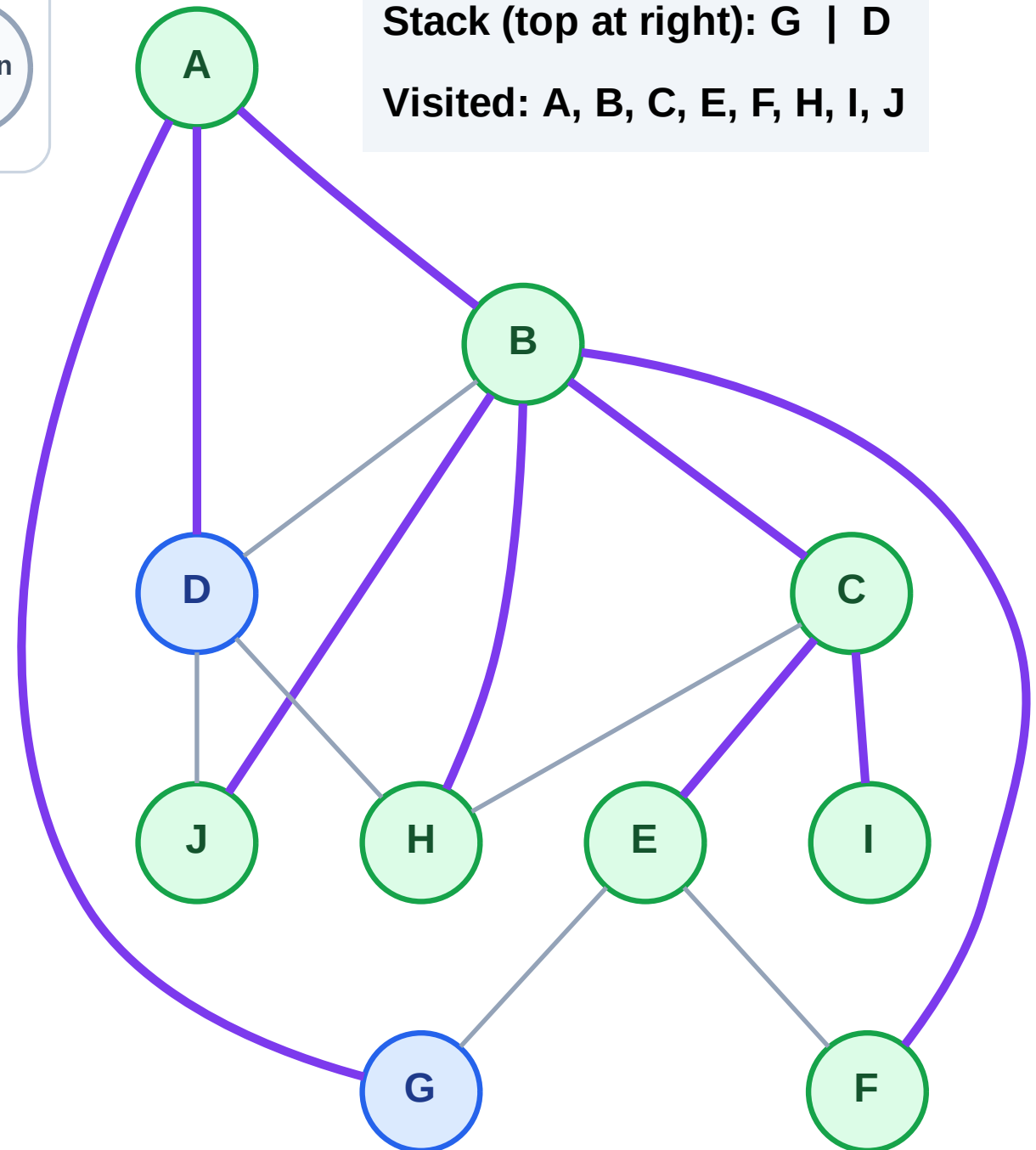
Step 17: No new neighbors from J

Legend



Stack (top at right): G | D

Visited: A, B, C, E, F, H, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

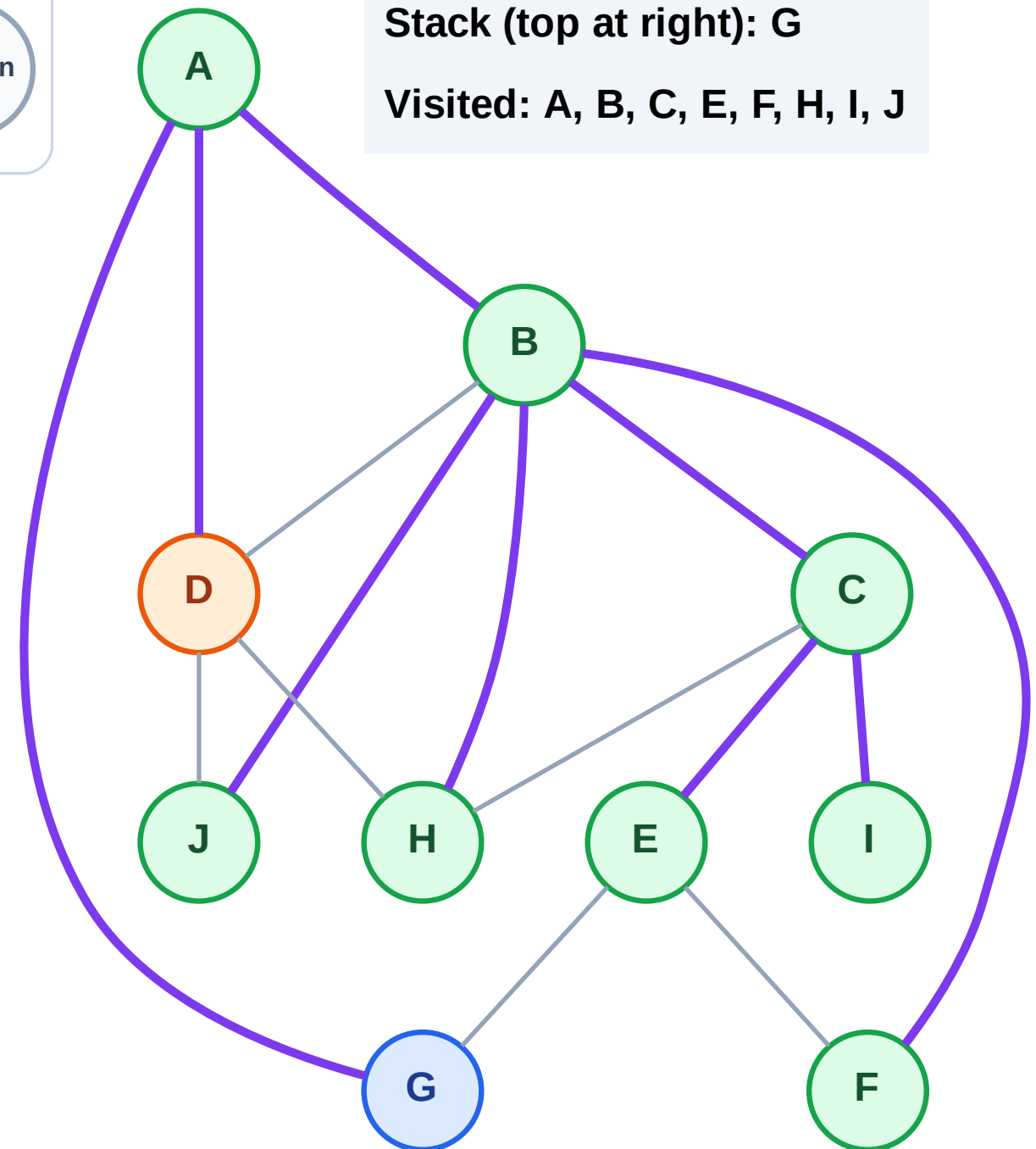
Processing

Discovered

Unseen

Stack (top at right): G

Visited: A, B, C, E, F, H, I, J



DFS Traversal

Step 19: No new neighbors from D

Legend

Complete

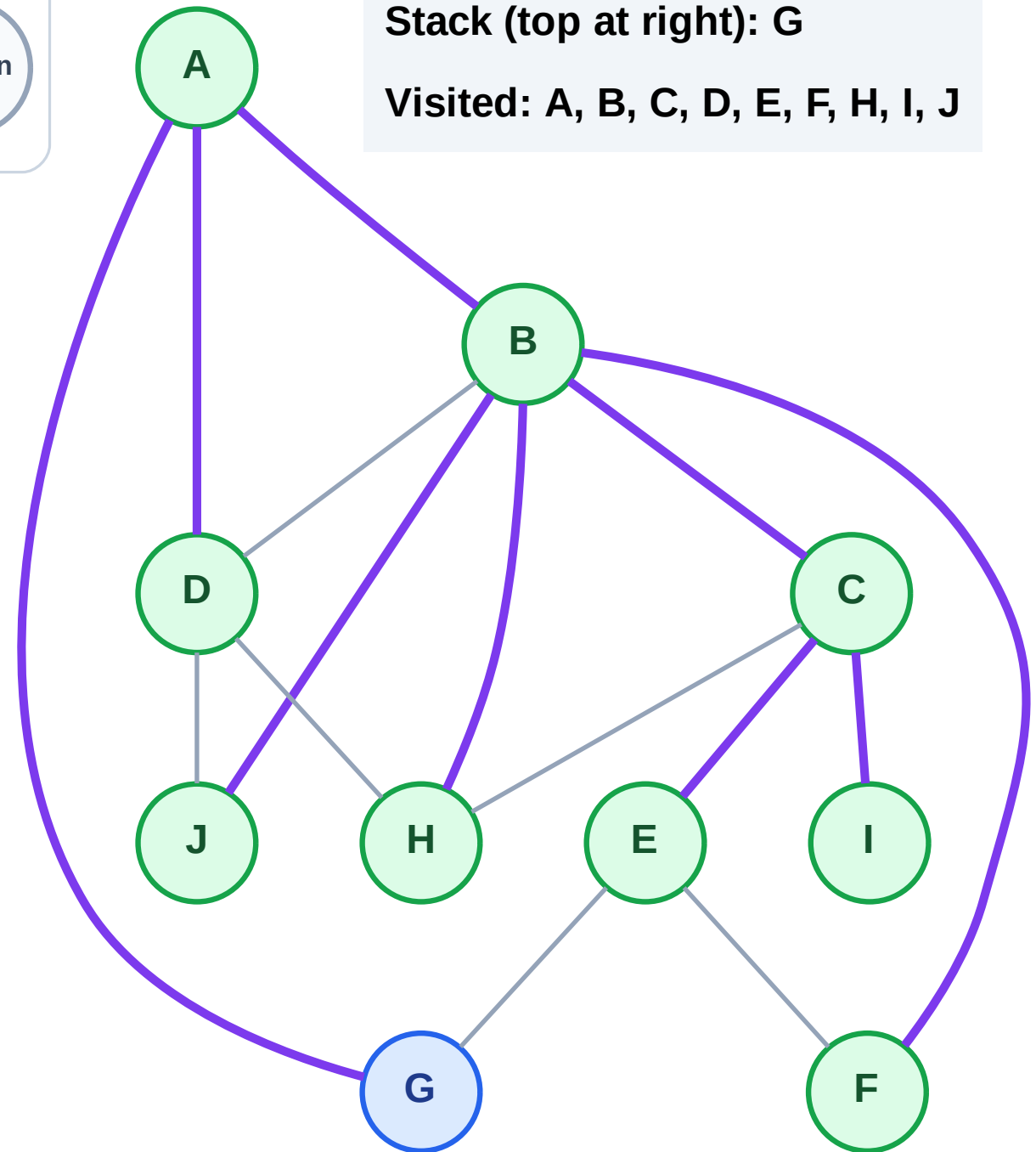
Processing

Discovered

Unseen

Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 20: Process node G

Legend

Complete

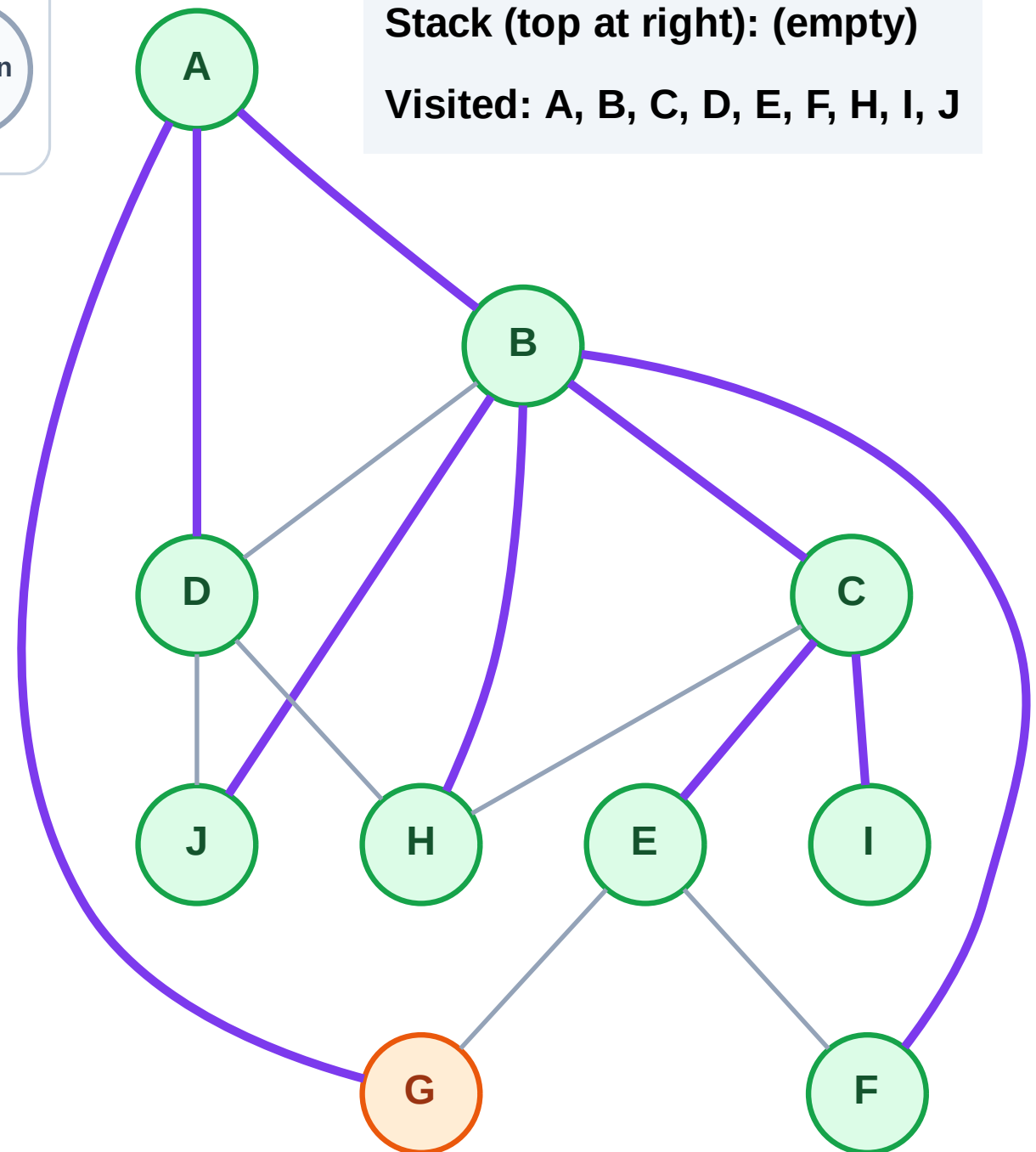
Processing

Discovered

Unseen

Stack (top at right): (empty)

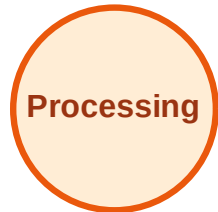
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

